The geometry of single-cell foundation models: what they inherit, what they add, and what shapes it

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Abstract

Single-cell foundation models are said to learn the geometry of cell state, and that geometry is increasingly used to justify their use for target prioritisation and perturbation prediction. We ask three questions of it across nine representations plus a model-free control: what does it look like, is any of it the model's own, and what makes it take the shape it does.

What they inherit. Cell-level geometry is a re-description of the input. Measured end to end on the cell cycle, full expression scores 0.929, the model's own tokenised input 0.882, and the model 0.875, so the tokeniser costs seven times what the transformer contributes. Against matched expression baselines the models win 1 of 45, 0 of 36 and 0 of 20 comparisons; raw data completes a deleted manifold region better than they do, and transfers across tissues better as well. Knowing this makes an accurate anatomy more useful, and several standing assumptions turn out to be wrong: the cell-cycle manifold is a *filled disk rather than a ring*, it is *flat*, its angle encodes phase while its *radius* encodes cycling strength, and it is *continuous* rather than a few discrete states.

What they add. The gene-token table holds facts absent from every individual input. MaxToki-1B places any of ~18,000 genes on one of 22 autosomes at 0.880 balanced accuracy (chance 0.045) under a 10-megabase neighbourhood holdout, against 0.720 for the strongest training-free factorisation of the same cells, and the model acts on it: steering a chromosome direction raises the model's own predictions for that chromosome's genes at untouched positions, positive for 132 of 132 chromosome × strength × seed combinations, and reproducing a measured leukaemia karyotype at $r = +0.43$. A cell-sentence model given one gene symbol and no other phase information places it correctly at AUROC 0.989, while an anagram of the same letters scores chance. This is a class of latent variable distinct from prior emergent-world-variable results, namely a property of the *vocabulary*, absent from any single input and orthogonal to the training objective.

What shapes it. On synthetic corpora with planted ground truth, a model allocates representational distance *where its own output distribution turns over fastest* (+0.160 over control, $p = 0.0041$, 8/8 seeds, localised to the manipulated arc in 5/5). Density does not predict where the distance goes, and difficulty predicts it with the sign reversed. This law predicts the one real-data geometry that is demonstrably the model's own: C2S gives extra representational room to the G1→S restriction point, the commitment step of the cell cycle. Navigability is fixed separately, by topology: since $\oint \mathbf{w} \cdot d\mathbf{x} = 0$, a constant steering direction must stall a quarter-turn into any closed manifold, which we verify analytically, synthetically, and in seven representations.

Running through all three is a methodological point we make explicit and supply tests for: a manifold visible inside a model is usually a re-parametrisation of one already present in its input, and telling the two apart requires a competitor built from the same cells rather than a permutation null.

1 Introduction

Foundation models trained on single-cell transcriptomes [1–4] are now proposed for decisions with real consequence: which gene to target, what a perturbation will do, what a cell in diseased tissue is. Much of the case for them rests on geometry. These models are said to organise cells along developmental trajectories, around the cell cycle, and into lineage hierarchies. If that geometry is the model's own, it is evidence of learned biology that can be read out, steered, and trusted.

The trouble is that geometry is easy to find and hard to attribute. The dominant statistical structure in transcriptomic data is that genes expressed together stay together, and almost any biological category one might wish to claim, including cell type, cell state, pathway and phase, is largely *defined* by co-expression. A representation that faithfully re-describes its input will therefore display all of that structure, and a probe will read it out at high accuracy, without the model having learned anything a linear method on the raw data could not supply.

A manifold in a model is usually a manifold in its data. Naming this failure precisely is one of the things this paper is for. A model can display a manifold for two quite different reasons. It may have built one, or it may be re-describing in new coordinates a manifold that was already in the data. From inside the model the two cases look identical: the same clean low-dimensional structure, the same high probe accuracy, the same interpretable axes, the same satisfying picture. Only one thing separates them, and it is not visible in the picture: whether a competitor holding the same information does as well. Reports of geometry in these models rarely run that comparison, so on the evidence usually given, a published manifold is a re-parametrisation of the input rather than a discovery about the model. The same worry has been raised from the language side: Prieto et al. [5] show that the circular arrangement of month words inside a model is recoverable by running PCA on the co-occurrence statistics of the text itself, so the geometry is inherited from the data's covariance. We treat the separation as a measurement problem, give the tests that settle it (Section 6), and apply them to every claim we make.

This paper separates three questions that the field routinely runs together, and answers each across nine representations. *What does the geometry look like?* We give an anatomy, done carefully enough to overturn several standing assumptions. *Is any of it the model's own?* We answer against baselines built from the same cells and, where possible, given exactly the model's own input. *What makes it take the shape it does?* We answer on synthetic corpora where we plant the answer, which lets us state a mechanism rather than a correlation. The organising finding is that the answer splits by *what is being represented.* A **cell's** state is determined by its own input, and the model re-describes it. A **gene's** meaning exists only across the corpus, and the model genuinely learns it. That boundary organises everything that follows.

Contributions.

1. **An anatomy of the cell-state manifold** (§3). It is a filled disk rather than a ring, it is flat, it decomposes into angle-as-phase and radius-as-cycling-strength, and it is continuous rather than discrete. Five independently trained models agree on the developmental axis at held-out cosine 0.98 to 0.998. All of this structure is present in raw expression.
2. **Vocabulary-level facts that no input contains, and evidence they are used** (§4): chromosome decoded genome-wide beyond a training-free factorisation and validated causally against an external karyotype; gene identity carrying cell-cycle phase with an identity-destroying control; function beyond a co-expression-matched null; and composable, steerable

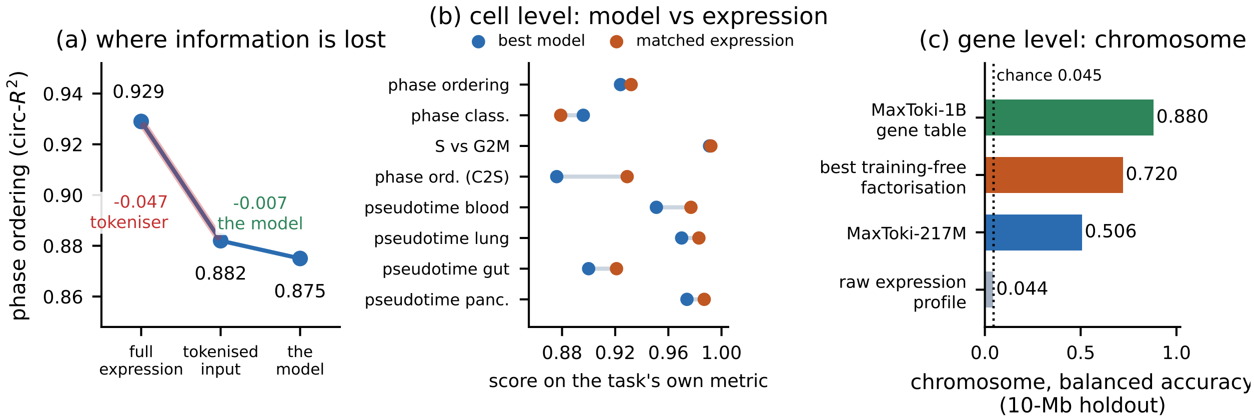

Figure 1: Two representational regimes. (a) On the cell cycle, information is lost at tokenisation, not in the model: the transformer contributes −0.007 against the tokeniser's −0.047. (b) Cell-level tasks, best model against a matched expression baseline. The two sit on top of each other on every task, and expression is ahead on seven of eight. (c) At the gene level the model holds a fact no single cell contains, beyond the strongest training-free factorisation of the same cells.

lineage-regulator geometry.

3. **A mechanism for how representational geometry is shaped** (§5): models spend distance where their own output changes fastest, demonstrated with planted ground truth and confirmed in a real model; a theorem that predicts when steering must fail and the operator that works when it can; and a measurement showing that the trajectory arrives bent at layer 0 rather than being built with depth, with 40% of what has been reported as curvature turning out to be vector magnitude.

4. **A protocol for attributing geometry to a model rather than to its input** (§6): six tests that separate a learned structure from a re-parametrised one, each stated so that it can be applied to a published claim without rerunning the original study.

2 Setup

2.1 Models and data

Nine representations plus a model-free control, on shared substrates: scGPT [1] (encoder, value-binned input; layer 11 and its native MVC head); Geneformer V1/V2 [2] (encoder, rank input); MaxToki 217M and 1B (Llama decoders, rank input, untied `lm_head`; all layers and both tables); STATE-SE/ST [6] and UCE-100M [3], whose gene tokens *are* frozen ESM-2 protein embeddings [7] and which therefore serve as the sequence control for vocabulary questions; C2S-Scale 2B and 27B [4, 8] (text LLMs over rank-ordered gene name sentences); and raw expression. Substrates: Replogle K562 and RPE1 non-targeting controls [9] (cell cycle, $n = 3{,}000$ each); Setty CD34$^+$ bone marrow [10], fetal gut, lung airway and mouse pancreas [11] (development); Tabula Sapiens [12] (cell type and gene context). Cell-cycle phase is scored with the standard marker sets [13, 14]. Full details in Supplement A.

2.2 Three questions

We distinguish **readable** (a probe recovers it), **model-specific** (it beats a competitor built from the same cells, given the model's own input where possible), and **causally used** (intervening changes the model's own output, read through its native head, not a fitted probe). These three are *independent*, and all combinations occur in our data (§6). They are commonly treated as a ladder, on which readable implies model-specific and model-specific implies used. No step of that ladder holds.

2.3 What kind of object each structure is

"Manifold" is used loosely for any low-dimensional structure. Most objects here are not manifolds, and the distinction decides which operations are possible at all. A *lookup* (chromosome: 22 unordered classes) admits no path, only signed saturating shifts. An *axis* (paralogy) admits a fixed direction. An *open tree* (pseudotime) needs a fixed direction plus projection onto the local data tangent. A *closed curve* (the cell cycle) makes a fixed direction provably impossible (§5.3). A *filled region* steers separately in angle and radius. A *fitted embedding* is a property of the fitting procedure, not of the model. We use "representational geometry" as the umbrella and name the type wherever we make a claim.

3 Cell-state geometry is inherited from the input

This section is about cells. §3 first asks who the cell-level geometry belongs to, the model or the data, and then describes that geometry on the two structures these models are most often said to build: the cell cycle and the developmental trajectory.

3.1 The cell-level representation is a re-description of the input

We start where the attribution question can be settled cleanly, because the answer changes how every later result about cells should be read.

What is being tested. Fix one task, one set of cells and one probe, then vary nothing but the representation the probe reads from. The task is to recover where a cell sits in the cell cycle: each of 3,000 K562 cells carries a phase angle computed from standard marker sets, and the probe is a linear readout scored by circular R^2, which is 1.0 for perfect recovery of the angle and 0 at chance. Give the probe the full expression vector and it scores what the data allows. Give it the model's internal state and it scores what the model preserves. The informative rows are the ones in between: representations holding exactly what the model was given and nothing more. C2S-Scale makes those rows constructible, because its input is written out explicitly as the top-512 gene names in rank order with magnitudes discarded, so a baseline can be handed *exactly* the model's own input. Reading the rows in order (Table 1) then separates the information lost when a cell is tokenised from the change produced by running the transformer over it.

The model sits *on* a linear readout of its own input (Table 1, Fig. 1a). This replicates under a frozen cross-cell-line transfer (K562 → RPE1), where the model scores 0.789 against 0.792 for a baseline matched on *both* encoding and gene-selection procedure, a paired $\Delta = -0.0022$, 95% CI $[-0.0144, +0.0105]$. Matching only the encoding gives a spurious "the model adds information" result ($\Delta = +0.0153$, CI excluding zero), because a cell sentence draws its top-512 from the cell's

Table 1: Information accounting on the cell cycle, end to end. Each row is a linear readout of cell-cycle phase from a different representation of the same 3,000 K562 cells. Reading down the table, the loss from full expression to the model's tokenised input is 0.047; the further loss from that input to the model is 0.007.

representation	phase ordering (circ-R^2)
full expression, 6,546 genes with values	**0.929**
top-512 genes with values	0.885
top-512 set, binary presence, *the model's input*	0.882
top-512 set + rank, *the model's input*	0.868
C2S-2B L21, the model	**0.875**

full panel while that baseline drew from the shared genes. Both must be matched (Supplement B.1). This is the protocol of §6 in miniature: the same experiment gives opposite verdicts depending on how the competitor is built.

Three further lines agree (Fig. 1b). On matched-dimensionality benchmarks the models beat expression in **1 of 45** cells and **0 of 36** (C2S); on developmental pseudotime against a *nonlinear* expression probe, **0 of 20**. Deleting a contiguous arc of a circular trajectory from training, the model bridges the hole, but raw expression bridges it better at every width, with the deficit growing from -0.004 at 30° to -0.027 at 120° (Supplement B.3). And a pseudotime probe fitted on one tissue and applied unchanged to another collapses for every representation: within-tissue Spearman 0.87 to 0.95, best cross-tissue 0.63, with 2 of the 3 pairs above 0.5 won by raw expression (Supplement B.4).

What follows from this. The cell-level geometry belongs to the data. An accurate description of it is still worth having, as a coordinate system on the expression manifold that is easier to compute with than the raw counts. It is not evidence that the model learned biology. The next two subsections do the describing, and §4 looks for the model's own contribution elsewhere.

3.2 The cell cycle: the single-cell version of the day-of-week circle

Language models place the days of the week on a circle, and place parts of the day on a second, orthogonal circle [15]. That result is the template for a widespread claim in single-cell biology, that these models likewise build a cell-cycle circle. The cell cycle is the one cellular process everyone agrees is periodic, and a genuinely circular internal variable would license a specific set of operations: read the phase, move along it, interpolate between phases. We asked what that object actually is. Three widely repeated descriptions of it turn out to be wrong.

It is a filled disk rather than a ring. A ring and a disk support different operations, so the difference is not cosmetic: on a ring, radius carries nothing and every cell is at the same distance from the centre; on a disk, radius is a second variable to read and to steer. Calibrated against synthetic ground truth (Fig. 2a), we report two statistics: the coefficient of variation of the radius, and the fraction of cells lying inside half the median radius. A planted ring gives 0.15 and 0.000; a planted disk gives 0.35 and 0.126. Observed: C2S-2B 0.38 and 0.108, C2S-27B 0.38 and 0.110, raw expression 0.42 and 0.122. All three match the disk. Persistent homology inside the fitted phase plane agrees ($H_1 = 0.227$ against a matched-covariance null 0.251 ± 0.063, $z = -0.4$). Earlier low-dimensionality statistics were computed on bin centroids, which describes the fitted summary, not the cells.

It is flat. Curvature is what would make a cell-cycle representation more than a rotated copy of a plane in the data, so its absence is worth establishing. The loop lies in a fixed linear 2-plane in every representation, and H_{flat} is never rejected (scGPT out-of-plane rotation 22.7° against a planar-circle null 25.6 ± 2.3, $p = 0.905$). Two consequences: a two-component linear readout loses nothing, so nonlinear embeddings of the cell cycle add only visual appeal; and the stall result of §5.3 cannot be blamed on a difficult ambient shape, because there is none. This is a statement about the cell-cycle loop alone. The *developmental* trajectory does bend, and §5.5 measures by how much.

Angle is phase; radius is cycling strength. This is where the day-of-week analogy pays off most directly. In language models the day-of-week circle sits orthogonal to a parts-of-day circle, so two different time variables occupy two clean, separable subspaces. The same two-factor structure holds here. Phase programmes load on the angle at $\sim$15× the rate of non-phase programmes, and cycling *strength* is the only covariate with meaningful radial loading ($R^2 = 0.414$ radial against 0.040 angular; Fig. 2b). A cell arrested in G1 and a cell cycling hard through G1 sit at the same angle and different radii. Anyone reading phase out of one of these models should read the radius too, because a low-radius cell has a phase estimate that means very little. Raw expression shows the same split more strongly (0.529), so this structure is inherited as well.

It is continuous, and here the analogy breaks in the cell cycle's favour. A week has seven days and nothing between them. The language-model structure is seven *discrete* points that happen to be arranged on a circle, so a state one third of the way from Tuesday to Wednesday has no referent, and the circle is a convenient layout for a categorical variable more than it is a manifold. A cell one third of the way from S to G2M is an ordinary cell. Whether the models treat it that way is a separate question, and the discrete answer is available to them: G1, S and G2M could sit as three attractors with jumps between them, which would make interpolation meaningless and steering a matter of hopping between basins. They do not. Three tests scored 0 (continuous) to 1 (discrete) give behavioural interpolation 0.17, metric profile 0.10 and occupancy 0.00 (Fig. 2c): no empty regions around the loop, no jump in the metric, and interpolating two cell states while reading the model's own marker logits moves the output gradually through intermediate phases, with the intermediate state a genuine mixture. The model is, if anything, *more* uniform around the loop than the biology.

That interpolation test is the one that exposes *activation plateaus* in language models, where moving from one input sequence to another leaves the output almost unchanged along most of the path before a sudden change part way through [16]. Those plateaus sharpen with depth and follow the input-output sensitivity of the MLPs. We ran the equivalent interpolation here and got the opposite on both counts: no plateau, and a plateau statistic that *weakens* with depth ($0.179 \rightarrow 0.115 \rightarrow 0.056$ across layers 9, 15 and 21). Supplement B.6 gives a falsifiable account, namely that a plateau needs a discrete output alphabet, and a graded distribution over hundreds of marker genes offers no cliff to form one. The two observations point the same way: the language-model feature is discrete and the cell-cycle coordinate is not.

So the cell-cycle object is closer to a manifold in the strict sense than the language-model structure that motivated looking for it. It is a continuous angular coordinate with no privileged points, lying in a plane, carrying a second continuous coordinate that is also meaningful. The day-of-week circle is a categorical variable wearing a circle's clothes.

Where the circle comes from. On both sides of the analogy it comes from the data. Prieto et al. [5] take the months of the year, show that month words carry cyclic co-occurrence statistics

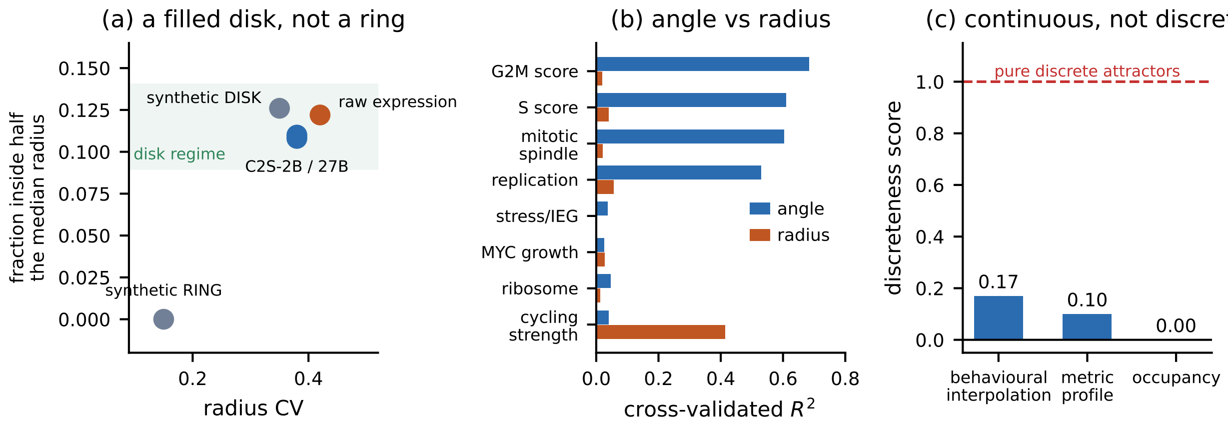

Figure 2: Anatomy of the cell-cycle manifold. (a) Against synthetic ring and disk calibrations, every representation, including raw expression, matches the disk. (b) A clean two-factor decomposition: phase programmes load on the angle, cycling strength on the radius. (c) Continuity scores; 1.0 would be discrete attractors.

in text, and recover the same circle from a PCA of those statistics as from a PCA of the learned weights; they call such structures inherited from data statistics. Every anatomical property above behaves the same way here. Raw expression matches the disk calibration (0.42 and 0.122), carries the angle-radius split more strongly than the models do (0.529 against 0.414), and supports the phase readout better (0.929 against 0.875). Two different domains, two different model families, and the same conclusion about where a reported circle comes from.

What the four results give a reader. Read together, they say the cell-cycle object is a flat, filled, continuous disk with two readable coordinates, and that it belongs to the expression data. That is a usable description: it says which readouts are meaningful (angle and radius, jointly), and it already forecasts the steering result of §5.3, because a closed loop is exactly the topology on which a constant steering direction is guaranteed to fail.

3.3 The developmental manifold: the object every steering claim rides on

The cell cycle is closed and periodic. Development is the other shape these models are asked to represent, and the shape with commercial consequences, because fate prediction, in-silico differentiation and target ranking are all operations on a developmental trajectory. Its geometry decides which of those operations can work at all, which is why we measure the shape before measuring the performance.

Pseudotime is decodable everywhere (full-space linear R^2 0.78 to 0.97 across 20 model × tissue cells). The local tangent *rotates* along the trajectory, with θ_{far} of 48 to 113° between the first and last pseudotime quintile, exceeding a heteroscedastic residual-permutation null in **16 of 16** cells. The object is therefore a branching tree of locally-linear segments, not one straight axis, and that single fact is what forces the operator of §5.4 to be local: a direction fitted globally points the wrong way by up to 113° at the far end of the trajectory it was fitted on.

Five independently trained models agree on the developmental axis, at held-out CCA-mapped gradient cosine 0.9804 to 0.9975 across 20 ordered pairs, against a random-rotation null at 0.36. Models trained by different groups on different corpora with different objectives converge on the same direction, which is strong evidence that the direction is a property of the biology they were all trained on. Local intrinsic dimension falls as cells mature ($\rho = -0.31$ to -0.75), a potency gradient, though a counts baseline reproduces it.

Pretraining does pay in one narrow place, which we state precisely because it is the exception to the rest of §3: endpoint targeting on hard, low-headroom trajectories (mouse pancreas endpoint purity $0.207 \rightarrow 0.569$, with 4 of 4 models beating expression; lung $0.599 \rightarrow 0.862$, 4 of 4), and making trajectories *linearly* readable, a real gain that dissolves against a nonlinear probe (Supplement B.5). The pancreas gain concentrates in the rare fates, which is where a pretrained prior would be expected to help.

What follows from this. The developmental object is a tree of locally-linear segments whose tangent turns by up to 113° from one end to the other, and five independently trained models agree on where it points. Anything that treats it as a single global direction is therefore mis-specified by construction, which is the failure §5.4 diagnoses and repairs. This closes the cell: the geometry is the data's, we now know its shape, and we know which operations that shape permits. §4 asks whether these models add anything at all, at a different level.

4 Gene-vocabulary facts the models add

§3 closed off the cell: the geometry is inherited, with the narrow exception of endpoint targeting on rare fates. The next question is whether there is any level at which these models add something outright.

Our measurements do not explain *why* the cell level came out that way, and we do not claim it had to. What they do is suggest where to look next. A cell's state turned out to be largely recoverable from that cell's own expression vector, and nothing the models did improved on that recovery. A gene's meaning differs in one concrete respect: it is not present in any single cell and can only be assembled across many of them, so a single-cell baseline has nothing to offer there. That makes the gene level the natural place to look for a contribution from pretraining. It is a hypothesis, not a deduction, and this section tests it. It holds.

4.1 Chromosome membership, genome-wide

We want a property that is unambiguously absent from every input, so that a positive result cannot be explained by re-description. A gene's chromosome is such a property. It appears in no cell's expression vector. It is recoverable only by aggregating co-occurrence across cells, because genes in the same chromosomal neighbourhood tend to be co-regulated. It is also irrelevant to the training objective, so nothing pushes a model to represent it.

Decoding. A 22-way classifier on the gene-embedding table, split over genes, under GroupKFold with **10-megabase genomic blocks** so that a gene's entire neighbourhood, including duplicates, tandem arrays and TADs, is held out together and the gene cannot be placed by recognising a near-identical twin: MaxToki-1B **0.880**, strongest training-free factorisation 0.720, MaxToki-217M 0.506, raw expression profile 0.044, chance 0.045 (Fig. 1c). Permuted labels give 0.045 and random-vector embeddings 0.042; removing HOX and the protocadherin array changes nothing; holding out whole token-ID blocks retains 96%, so this is not tokeniser adjacency; embedding norm alone predicts at chance. ESM-2 reaches 0.19 on a random split but **collapses to** 0.07 under the holdout (Fig. 3a), because its signal was tandem-duplicate family resemblance. That collapse is the natural experiment separating sequence from locus, and it is why the sequence control has to be run under

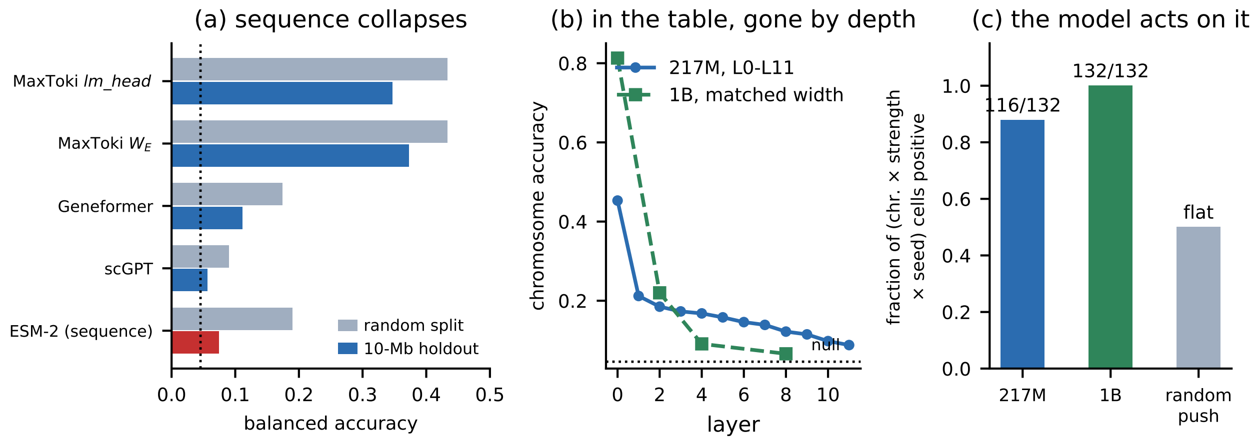

Figure 3: Chromosome membership. (a) Under a 10-Mb neighbourhood holdout the model retains 78 to 85% of its random-split accuracy, Geneformer 51%, scGPT 24%, and the ESM-2 sequence control collapses to 20%. (b) The variable lives in the table and the earliest layers. (c) Steering is positive for every chromosome, strength and seed in the 1B.

the same holdout as the model.

Two limits. Only the 1B clears the factorisation; the 217M falls short of it. And on a *label-free* retrieval test over gene pairs $\geq$20 Mb apart the ranking **inverts**, with the factorisation scoring 0.565 against both models' 0.529. The model's margin is what a supervised probe extracts; the raw geometry is no more accessible.

Causal use. Readability alone would leave this a curiosity, so we asked whether the model's own computation uses the variable. Adding a chromosome-C direction to a random half of a cell's gene tokens raises the model's own predicted probability of chromosome-C genes at the *other, untouched* half. Positive for **132 of 132** chromosome $\times$ strength $\times$ seed combinations in the 1B (116/132 in the 217M); a norm-matched random push is flat; the response is dose-monotone (Fig. 3c). This is read through the model's native next-gene logits rather than a fitted probe. **External validation:** the same directions reproduce a measured aneuploid karyotype's chromosome-level signature at $r = +0.43$, against a shuffled-karyotype control at $\approx$0.

Where the variable lives. The variable is strong in the vocabulary tables (0.516 output, 0.453 input) and decays monotonically to 0.088 by the last layer with no rebound; the 1B shows the same pattern more sharply (0.813 $\rightarrow$ 0.066 at matched width, retaining 8%; Fig. 3b). Steering works at the input embedding (+0.0801, 22/22) and is indistinguishable from zero by layer 5. *A variable can be functionally important while remaining undetectable as a feature at depth*, which is a caution for any layer-sweep study that reports the last layer only. We also tested the mechanism usually assumed for this, attention between same-chromosome genes, and the data rule it out: across 11 layers $\times$ 8 heads, same-chromosome gene pairs show **no attention enrichment** once rank-distance is matched, with 0 of 88 heads significant and the strongest head changing identity and flipping sign between samples. The causal effect is real; the route runs through the embedding table, not through chromosome-sorted attention (Supplement C.1).

4.2 A gene's identity carries its cell-cycle phase

Chromosome shows that a corpus-level fact can sit in the table. The obvious follow-up is whether such facts are ever *biological*, or whether the table only holds bookkeeping. Here the test is the strongest available for locating knowledge in the weights rather than the prompt, because the prompt is stripped of the answer. Take a real cell, remove *every* cell-cycle gene so the input carries no phase covariance, insert **one gene symbol** at a fixed rank, and read which pole of the real-cell phase circle the state lands on (Fig. 4). Human `CCNB1` scores AUROC **0.989** ($p = 0.0002$, 40.7° separation), while the **anagram** `NBCC1` scores chance (0.511, $p = 0.43$). The effect survives residualising on symbol length, abundance and detection rate (0.908, $p = 5 \times 10^{-5}$) and holds for 30 of 31 exact length-matched pairs ($p = 3 \times 10^{-8}$).

Scope. What transfers from the day-of-week result in language models is the *shape of the evidence*: a minimal prompt, an identity-destroying control, and structure that survives both. The strength of the claim does not transfer. This is a two-class axis with a compressed 40.7° arc, offset from where real cells sit. "The model knows `CCNB1` is mitotic" is supported. "The model has an internal clock it can place any gene on" goes beyond what we measured.

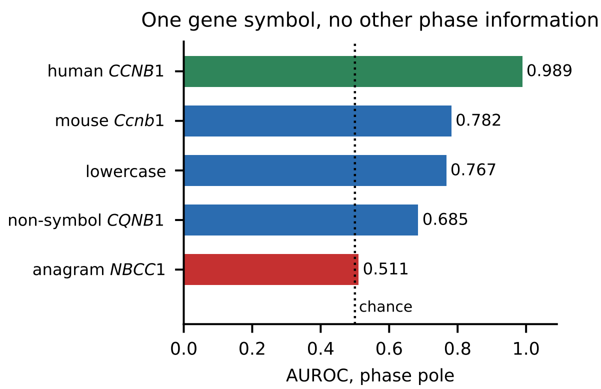

Figure 4: A surface-form ladder. Five inputs, identical except for the single gene symbol inserted: human `CCNB1` 0.989, mouse `Ccnb1` 0.782, lowercase 0.767, the non-symbol `CQNB1` 0.685, and the anagram `NBCC1` 0.511. Accuracy falls monotonically as the symbol moves away from a real human gene name, and the anagram, which preserves every letter, sits at chance.

4.3 Function, paralogy, and lineage switches

If the vocabulary table is a genuine store of corpus-level knowledge, the knowledge in it should not stop at chromosome and phase. Three further probes ask how far it goes, and each is paired with the baseline that could explain it away.

Function beyond co-expression. In C2S-Scale, same-function gene pairs are closer than different-function pairs *within matched co-expression and abundance bins*: $\Delta\cos$ of 0.044 to 0.062, $z = 3.9$ to 6.6 against 200 co-expression-matched label permutations. This is one of very few results here that clears a co-expression-matched null, not merely a permutation null, which is the distinction §6 is about.

Paralogy as a composable coordinate. Leaving one entire HOX cluster out, the paralog coordinate transfers to it at held-out $\rho = +0.833$, with the correct sign in 4 of 4 clusters ($p = 0.005$). Word2vec-style arithmetic works: `HOXA9` − `HOXA1` + `HOXB1` lands on `HOXB9` at 3.95× a strict within-cluster null ($z = +13.5$), where co-expression fails entirely. *Caveat:* protein sequence recovers paralogy too, so this tells us what the model learned, not something only the model knows.

Antagonistic regulators sit antipodally, and are steerable. RORC/FOXP3, GATA1/SPI1 and PAX5/PRDM1 sit on opposite sides of a marker-defined subspace, beating *both* co-expression and sequence on 3 of 5 pairs (GATA1/SPI1 separation 0.818, $p < 10^{-4}$; subspace cosine -0.394 for the model against $+0.007$ for co-expression and $+0.699$ for ESM-2). Pushing toward one pole raises that lineage's held-out markers *and suppresses the other*, the signature of a bistable switch, at $+4.211$ $[+3.892, +4.539]$ for GATA1/SPI1 and $+5.565$ for PAX5/PRDM1, read through the model's own logits. Off-tissue pairs reproducibly reverse, which is itself informative: the geometry is gated by context, not fixed.

4.4 The class of latent variable chromosome belongs to

The chromosome result belongs to a different family from the emergent-world-variable literature, and the difference suggests where to look next in other domains.

Every prior emergent-world-variable result concerns something *dynamic and determined by the current input.* An Othello board is a function of the moves [17, 18]. A chess model's board state and player skill are recoverable from the position [19, 20]. In language models the latitude and longitude of a place, or the date of an event, are recoverable [21], but the place or event is *named in the text being processed.*

A gene's chromosome is different in kind. It is **absent from every single input** and **orthogonal to the training objective**. It is a fixed property of the *vocabulary*, recoverable only from cross-corpus co-occurrence, and it is read from a frozen table rather than from context-dependent activations. At the developmental HOX clusters its causal use is moreover decoupled from the current input. The generalisation we propose: **look for latent variables that are properties of the token vocabulary rather than of the current sequence.** Any domain with a fixed vocabulary and corpus-level structure over it, including genes, chemical fragments, protein domains and code identifiers, should have them.

5 What shapes the geometry

So far the paper has said whose geometry these structures are. This section asks what determines their shape, and answers in three parts. First, two questions about what a training objective does to a representation: when does a model modify a metric it inherited (§5.1), and what is a token table for (§5.2)? Both are settled on synthetic corpora where the ground truth is planted rather than inferred, run through a standard transformer with a published input encoding. Second, two questions about what the resulting shape permits: which steering rules a closed manifold forbids (§5.3), and which rule works on a tree (§5.4). Third, where the shape comes from in the stack at all (§5.5), which turns out not to be where the usual account puts it.

5.1 Models spend distance where their own output changes fastest

Exactly one geometric property in our real-data corpus survived the attribution tests of §6 as the model's own, and we have not presented it yet because it needs the mechanism that follows. C2S *stretches* the cell-cycle metric at the G1→S restriction point: knot-gap CV 0.318 (2B) and 0.295 (27B) against raw expression's 0.193, with the largest gaps at a different and biologically decisive place from the data's own, replicating across a 13× parameter gap.

This is the paper's one clear case of a model doing something to a geometry instead of passing it through, and it is a third kind of thing. Prieto et al. [5] separate geometry that comes from correlations in the data from *value-coding* geometry that a task creates even when the inputs are uncorrelated. What we see here is neither, and sits between them: the *shape* is inherited, and the *metric on that shape* is task-driven. It is worth saying what the model did. It moved resolution to the restriction point, the step at which a cell commits to divide. That is not where the data's own largest gaps sit, and it is not an arbitrary place: the model gives extra representational room to the part of the cycle where the cell's decision is actually made. Whatever these models fail to add at the cell level, here one of them reshapes an inherited metric, and puts the reshaping somewhere biologically sensible. A single observation in one model family cannot say what triggers that.

We therefore planted the answer. Generate a circle with a perfectly *even* metric, then break exactly one thing inside one 90° arc, and ask whether the model's copy of that circle stretches there. The statistic is (model stretch − data stretch), so only a model-specific effect registers. Four corpora, four planted causes (Table 2).

Table 2: Four planted causes, one effect. Each row is a separate synthetic corpus in which one property was changed inside a 90° arc of a planted circle. Standard transformer, published input encoding, 8 seeds per arm. Only the output-change manipulation separates from control, and prediction difficulty moves the metric the other way.

what was changed inside the arc	model − data	vs control
control: nothing	$+0.027 \pm 0.095$	reference
the model's own output turns over fastest there	$\mathbf{+0.187 \pm 0.091}$	**+0.160**, $t = 3.43$, $\mathbf{p = 0.0041}$, 8/8
3× more cells sit there	$+0.097 \pm 0.039$	not separable
cells there are harder to predict	$\mathbf{-0.161 \pm 0.080}$	negative; sign against

The **localisation** is unambiguous: the model's largest gap falls *inside the manipulated arc in 5 of 5 seeds* for the output-change arm, against a control that scatters around the ring (Fig. 5). A confound acting on magnitude could not produce that; this is the model putting the warp in the right place, seed after seed. So **a model allocates representational distance in proportion to how fast its own output distribution changes**. Density does not predict it, and difficulty predicts it with the sign reversed. This is the mechanism behind the real-data observation, because the restriction point is precisely where the transcriptional programme turns over fastest. It also gives a way to predict, before looking, where a model will spend resolution on any manifold: find where its own predictions change fastest.

5.2 The gene table is a genuine map, and the model depends on it

The same synthetic setting answers the other open question, which is what a token table is *for*, and whether the chromosome result of §4 should have been expected. Assign every gene an arbitrary group label appearing in *no single cell*, the synthetic analogue of a chromosome, and sweep how strongly group membership drives co-occurrence. The table tracks a training-free factorisation across the whole dose-response curve, reaching 0.426 at full strength against the factorisation's 0.497 and a chance rate of 0.050. It **generalises to genes that never co-occur in any cell**, which is the synthetic form of the neighbourhood holdout, at four times the factorisation's own generalisation (held-out cosine +0.0147 against +0.0037; Stouffer $z = +20.2$). And **freezing the table at random initialisation costs a quarter of the model's performance** (val_corr $0.853 \rightarrow 0.612$), so the structure is load-bearing. Together these make the real-model chromosome

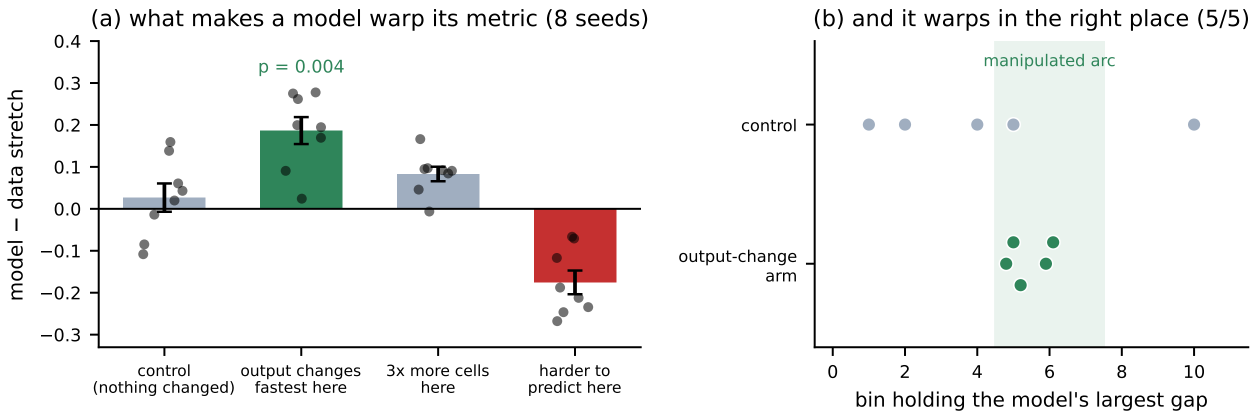

Figure 5: What makes a model warp an inherited metric. (a) Only the output-change manipulation separates from control; difficulty produces the opposite sign. Points are individual seeds. (b) The warp lands inside the manipulated arc in every seed.

result expected: a token table is where a transformer puts corpus-level structure, because it is the only component that sees the corpus and not a single cell.

5.3 Navigability is decided by topology, not by the model

The anatomy of §3 said the cell cycle is a closed loop. That fact alone, before any question about what the model learned, determines whether the most common steering method can work.

For any closed curve and any fixed direction $\mathbf{w}$, $\oint \mathbf{w} \cdot d\mathbf{x} = \mathbf{w} \cdot \oint d\mathbf{x} = 0$. A constant-direction steerer therefore does exactly zero net work per lap. Because the operator unit-normalises each step, a projected constant direction becomes $\operatorname{sign}(\mathbf{w} \cdot \hat{t})\,\hat{t}$: it advances while $\mathbf{w} \cdot \hat{t} > 0$, retreats once $\mathbf{w} \cdot \hat{t} < 0$, and converges to a stall where $\mathbf{w} \perp \hat{t}$, about a quarter-turn in, *even on a perfectly flat circle.*

Pre-registered, then verified three ways (Fig. 6). Analytically, above. On a synthetic zero-curvature circle, stall at 1.49 rad against the predicted $\pi/2 = 1.571$, with the pre-normalisation step norm collapsing to 0.09 of its initial value, a judge-free measurement of $\mathbf{w} \perp \hat{t}$. And in seven representations including raw expression with no model at all, fixed direction 0.01 to 0.36 laps against local-tangent 4.5 to 6.0. Measured cell by cell, the phase rate peaks *exactly* at the perpendicular crossing (90.7°), banking 55.3% of total advance before it. This single fact explains a family of failed steering results at a stroke, and it explains them without reference to any property of any model: the geometry of the target, rather than the quality of the representation, was the obstacle.

5.4 A steering operator that works

Diagnosing why steering fails is only half of it; the geometry also says what to do instead. The tree shape of §3 says the direction must be local, and the stall theorem says the step must be projected onto the tangent.

Local direction + tangent projection + retraction. On a branching trajectory, keeping one global direction but re-projecting it at every step onto the top-m right singular vectors of the $k = 100$ nearest real training cells lifts constrained advance from 0.461 to **1.239**, positive in **16 of 16** model × tissue cells and beating an oracle allowed to use held-out labels in 12 of 16. Ablations separate the ingredients: projection alone gives +0.778, local direction alone gives −0.072. On a

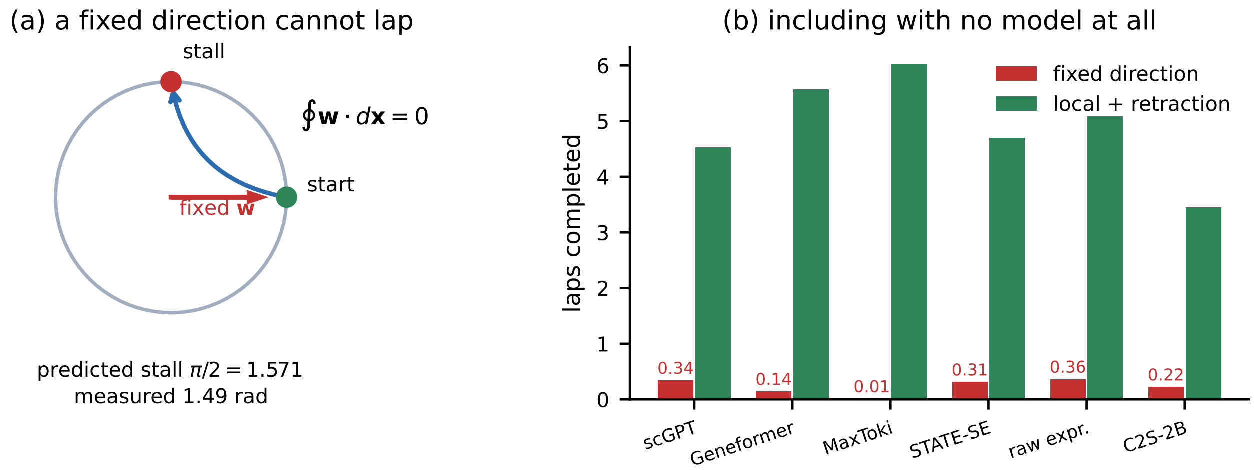

Figure 6: Topology, not the model, decides navigability. (a) A fixed direction does zero net work per lap and stalls a quarter-turn in. (b) The prediction holds in every representation, including raw expression.

closed loop the balance reverses and locality dominates, which is what the two topologies predict. Retraction cures outward drift but never the stall. The operator needs only a cell-by-gene matrix, so we release it as a model-free tool.

Behaviour follows geometry. A fitted manifold could be an artefact of the fitting procedure, so the last check reads the model, not the embedding. Walking a cell state around the fitted manifold and patching it into a real forward pass drives the model's *own next-gene predictions* around the cycle: 1.579 behavioural laps for the local-tangent rule against 0.212 for a fixed direction, 12 of 12 cells favouring local-tangent ($p = 2.4 \times 10^{-4}$), with a representation→behaviour isometry of 0.73 to 0.88. The geometry we fitted is the geometry the model computes with.

5.5 The geometry arrives bent, and part of the bending is size

One question remains about where geometry comes from: whether the network builds it with depth, as the usual framing assumes. It arrives at layer 0 instead, and a large part of what has been measured as curvature turns out to be vector magnitude. This subsection concerns the *developmental* trajectory. The cell-cycle loop of §3 is flat, and there is no conflict: a curve can bend through a high-dimensional space while a loop lies in a plane, and these are two different objects measured on two different substrates.

Whether these manifolds actually *bend*, as opposed to a quadratic term merely predicting pseudotime better than a linear one, was not established before. We separate **angular** curvature (real bending) from **radial** curvature (nonlinearity in vector magnitude, that is, LayerNorm and scale effects), at matched width across all layers (Table 3, Fig. 7).

The manifold does bend, since angular curvature clears its null in all three models. **But roughly 40% of the apparent curvature is magnitude rather than shape** in two of three, where the vector norm itself tracks pseudotime at +0.42 and +0.47; only the largest model is nearly pure. Any statistic that fails to separate the two overstates the geometry. And in all three, curvature **peaks at layer 0 or 1** and decays with depth, so the manifold *arrives bent* rather than being constructed. This sits in tension with the common framing of these representations as something the network builds with depth, and it is consistent with the rest of the paper: the structure comes in with the data and the embedding table, not from the stack.

Table 3: **Curvature separated into shape and size.** Angular curvature is genuine bending of the trajectory; radial curvature is nonlinearity in vector magnitude. The ratio column is radial/angular. "Peak layer" is the layer at which angular curvature is highest, out of the model's total; in all three it is the input or the layer next to it. The final column shows how strongly the vector norm alone tracks pseudotime.

model	peak layer	angular	radial	ratio	corr($\|x\|$, pseudotime)
scGPT	**0** of 11	+0.0480	+0.0209	0.44	+0.42
MaxToki-217M	**1** of 11	+0.0598	+0.0252	0.42	+0.47
MaxToki-1B	**0** of 20	+0.0552	+0.0050	**0.09**	+0.21

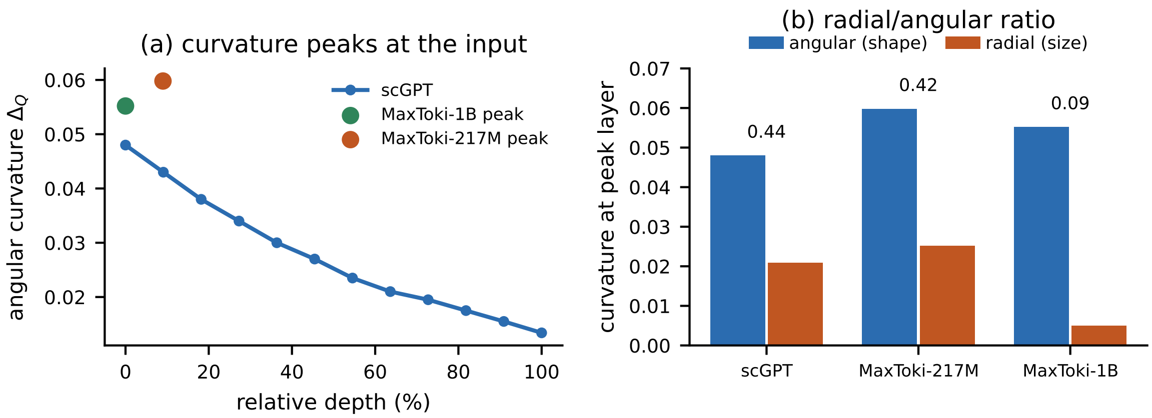

Figure 7: Curvature is real, partly radial, and front-loaded. (a) Angular curvature peaks at or next to the input in all three models. (b) Two of three are substantially radial, so a statistic that does not separate shape from size overstates the geometry.

6 Separating a model's geometry from its input's geometry

The results above rest on a small set of tests that decide attribution. We state them here as a protocol, because they are the part of this work most reusable by others, and because each can be applied to a published claim without rerunning the original study. Each test is paired below with a measurement showing what it catches.

1. Match the baseline to the model's input, on both encoding and selection. A baseline is only a competitor if it holds the same information. In the cross-cell-line transfer of §3, matching the encoding alone yields "the model adds information" ($\Delta = +0.0153$, CI excluding zero), while additionally matching how genes were selected yields parity ($\Delta = -0.0022$) on the same cells.

2. A permutation null says an effect is not noise; only a baseline says it is the model. This is the most consequential test and the one most often skipped. In **six** separate cases here a large z against a permutation null goes to approximately zero against a competitor built from the same cells (Supplement F.3). The most instructive: a label constructed from *pure sampling noise* scored 0.447 at $z = 35.8$, higher than the real biological label at 0.357, $z = 24.9$. A permutation null asks whether the labels carry the effect. Attribution asks whether the model does, and only a competitor can answer that.

3. The baseline panel must express the biology in question. A baseline built on cells that do not exhibit the phenomenon measures the panel's blindness rather than the model's skill. HOX genes are expressed in 3% of Tabula Sapiens cells, where a co-expression baseline scored 0.08. The trap runs in reverse too. On non-proliferating tissue the models score 0.767 on the cell cycle against a co-expression baseline's 0.495; rebuild the same baseline on cycling cells and it gives 0.879 against MaxToki's 0.358. The panel, not the model, moved.

4. Make comparisons invariant to orientation, magnitude and width. Three nuisance quantities masquerade as structure. *Orientation*: in cross-tissue transfer an apparent model advantage of +0.278 ($p = 0.016$) falls to +0.094 ($p = 0.23$) once the comparison is made invariant to the arbitrary sign of each tissue's pseudotime; the advantage was consistency of direction, not structure. *Magnitude*: roughly 40% of measured curvature is vector norm (§5). *Width*: comparing representations at native dimensionality measures width as much as content, and reducing MaxToki-1B from 2,304 to 256 components costs nothing at layer 0 (0.910 $\rightarrow$ 0.813) while collapsing layer 8 (0.265 $\rightarrow$ 0.066).

5. Score two nulls against each other before reporting an effect against one. If an estimator is biased, a null-versus-null comparison reveals it while a null-versus-real comparison hides it. One estimator here returns -0.489 comparing two structureless labellings against each other, larger in magnitude than the -0.227 it returns on the real comparison, so its output on the real comparison carries no information and the arm is not reportable (Supplement D.4).

6. Report readable, model-specific and used separately. Each of the three can hold without the others, so none of them implies another, and our data contain a worked case of every dissociation we could measure. *Used but not model-specific:* a functional-context axis is causally used at $p = 4.3 \times 10^{-7}$ (signed swing +0.815, 28 of 30 cells, dose-monotone) yet fails its co-expression-coherence null at $+1.16\sigma$. *Readable but not used:* sub-chromosomal position decodes at $\rho = +0.396$ in 22 of 22 chromosomes and beats both baselines, yet the causal response is flat along the chromosome. *Readable but not traversable:* cell-cycle phase reads at 0.93 from gene tokens while the geometry is a two-pole axis with minimum angular gap 1.6 to 30.7°, against 90° for a real circle.

A note on extraction. Two data defects, both found here, both biasing in a known direction, are worth flagging for anyone using the same public assets. Feeding raw counts to a checkpoint configured for binned input *understates the model and overstates its curvature*: correcting it raises pseudotime decodability by 0.109 and linear fit by 0.210, and *lowers* the measured curvature gap by 0.075. And in one widely used cached developmental dataset, differentiated cells sit *below* stem cells in pseudotime.

These tests are not gentle, and the first result we put through them was one of our own. A previously reported developmental representation extracted from a frozen model [22], benchmarked against scVI [23], Palantir [10], DPT [24], CellTypist [25], PCA and raw expression across 88 donor-holdout splits, does not survive them: the exported operator performs the same as a *random matrix on the same support* at every bottleneck width $\geq$10, and the model's own frozen representations lose to a dimension-matched *random projection of raw genes* by 0.0507 AUROC on **14 of 14** donors ($p = 0.0001$). Every check in the original was passed; none of those checks was a matched competitor. Supplement E gives the full battery.

7 Related work

Three literatures meet here: what latent variables neural networks build, what geometry has been claimed for biological foundation models, and how those models fare against simple baselines. Our position in each follows from the split of §3 and §4.

Emergent world variables. Board state in Othello [17, 18] and chess [19], look-ahead [20], and space and time in language models [21] establish that models trained on narrow objectives build representations of structure they were never taught; the form such variables take, including the circular features that motivate §3, is itself an active question [15, 26], as is how sharply a model's output responds to movement between inputs [16]. All of these variables are dynamic and determined by the current input, and §4 argues that vocabulary-level facts are a distinct class.

Where a model's geometry comes from. The closest work to ours is Prieto et al. [5], and our results are best read as building on it. They study superposition in a controlled autoencoder over bag-of-words text and show that when features are correlated the interference between them can be constructive: the model arranges features by co-activation, which produces semantic clusters and the cyclical structures reported in real language models. Their months-of-the-year circle is recovered from a PCA of the data as well as from a PCA of the weights, so the circle is inherited from the corpus covariance. We reach the same verdict about a periodic structure in a different domain (§3), and extend it three ways. First, the evidence is a competition instead of a correspondence: our baselines are given *exactly* the model's own input, which the explicit tokenisation of a cell-sentence model makes constructible, and we turn that comparison into a protocol (§6). Second, we add interventions read through the model's native head and an externally measured ground truth, neither of which is available for text. Third, two of our findings fall outside their two categories of correlation-driven and task-driven geometry: a metric that a model reshapes on top of an inherited shape (§5.1), and facts that no single input contains at all (§4). Their closing question, how to separate these effects in realistic settings, is the question §6 answers.

Geometry in biological foundation models. Low-dimensional cell-state structure is reported by the model papers themselves [1–3] and analysed spectrally in our earlier work [27], generally as encoding results without a causal test, and generally without a competitor built from the same data. Reports of gene embeddings reflecting genomic organisation are local, pairwise and correlational, without neighbourhood holdout or intervention. Our reading of that literature is the one stated in §6: absent a matched competitor, such reports characterise the data as much as the model.

Evaluation of single-cell foundation models. Zero-shot studies question how much these models add over simple baselines [28, 29], linear baselines match deep models on perturbation prediction [30], benchmark rankings are unstable under protocol choices [31], attention-derived scores largely re-encode co-expression [32], and a protein language model already supplies part of what single-cell pretraining is credited with [33]. That baselines are competitive is already established. Our contribution is the split: baselines win on cell state, the model wins outright on vocabulary facts. To that we add a mechanism for the geometry and a protocol for attributing it.

Factorisation baselines. Skip-gram is implicitly a factorisation of a shifted PMI matrix [34], so linear recoverability from a co-occurrence embedding is not by itself evidence of going beyond co-occurrence. Our label-free retrieval result is consistent with that caution.

8 Discussion

The boundary. A model can only add information its input does not already determine. In single-cell models that boundary falls between the cell and the gene, and it is sharp. Cell-level geometry loses to matched baselines on every axis we could measure, while gene-level tables carry facts no cell contains and act on them. The three parts of this paper are one argument: the boundary explains which geometries are inherited, the warp law explains what a model does to an inherited geometry when it does modify one, and the topology results explain what can be done with either.

In practice. Cell embeddings should not be expected to outperform careful expression baselines, and where they are used, the baseline must be given the model's own input. Gene-embedding tables are the component worth mining, and the neighbourhood holdout is the control that makes such claims credible. Because these tables encode chromosome, any comparison between groups differing in karyotype, such as tumour against normal or aneuploid disease states, inherits a confound that should be regressed out rather than assumed absent.

Limitations. The chromosome result clears its strongest baseline only in the larger of two checkpoints, inverts under a label-free test, and the baseline's own score varies sixfold across tissue panels for reasons we have not explained; the decisive control, scoring against Hi-C compartments and replication timing, is not yet run. Cross-tissue transfer is measured on six usable pairs. The synthetic work uses small models on simple planted structures. Attention was measured for chromosome but not for the cell cycle. The two extraction defects above are documented but not yet corrected across all affected analyses.

Single-cell models as a test bed for questions about language models. The questions asked here are not specific to biology. How a low-dimensional geometry forms in a trained network, when a model reshapes a metric it inherited, what a token table stores, and which steering operations a topology permits were all asked of language models first, and are hard to settle there. A single-cell model is a smaller and more tractable place to ask them, for three reasons that recur throughout this paper. Its input is a vector of numbers, so a competitor holding *exactly* the model's own input can be constructed, which is what made the accounting in §3 possible and is rarely available for text. Its vocabulary is fixed, finite and externally annotated, so a claim about what a token embedding knows can be checked against a genome instead of against intuition. And its ground truth is independently measurable: the cell cycle is periodic, genes sit where the assembly says they sit, and a karyotype can be measured without reference to any model. Several results here have direct counterparts in the language-model literature, including a circular representation of a periodic variable, a latent world variable that no single input contains, and the failure of a fixed steering direction on a closed feature. Where the counterparts differ they differ informatively, as with the continuity result of §3. Prieto et al. [5] close by asking how to separate data-driven from task-driven geometry in more realistic settings than a controlled autoencoder allows; a single-cell model is such a setting, and §6 is our answer to that question. We suggest treating these models as a controlled place in which claims about representational geometry can be tested more cheaply and more decisively, with the conclusions carried back.

Code and data. All analysis code, result files, figure scripts and the full control battery are released with the paper.

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Supplementary Material

Companion to "The geometry of single-cell foundation models: what they inherit, what they add, and what shapes it".

Contents: **A** Models, data and extraction · **B** Cell-state geometry in full · **C** Vocabulary-level facts in full · **D** Synthetic experiments in full · **E** A worked negative: the extracted-operator battery · **F** Retractions, instrument failures and things that did not work

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A. Models, data and extraction

A.1 Representations

model	architecture	input encoding	what we read	width
scGPT [1]	encoder, 12 layers, 8 heads	51 quantile value bins	L11 residual, mean-pooled; native MVC head	512
Geneformer V1/V2 [2]	BERT encoder	expression rank	`hidden_states[12]`	256 / 768
MaxToki-217M	Llama, 11 blocks, 8 heads	expression rank	L0–L11; `embed_tokens`; untied `lm_head`	1232
MaxToki-1B	Llama, 20 blocks	expression rank	L0–L20; both tables	2304
STATE-SE / STATE-ST [6]	bidirectional	ESM-2 gene tokens	L11, skipping CLS	2048
UCE-100M [3]	transformer	ESM-2 gene tokens	L2	n/a
C2S-Scale-2B / 27B [4, 8]	Gemma-2 text LLM	rank-ordered gene-name sentence, top-512, magnitudes discarded	last-token residual, L09/15/18/21/25 and L44	2304 / 4608
raw expression	n/a	log1p(CP10k)	PCA	varies
ESM-2 [7]	protein LM	amino-acid sequence	frozen gene embeddings	5120

Note on gene tokens. UCE's and STATE-SE's gene tokens *are* frozen ESM-2 protein embeddings [7]. They therefore cannot be asked questions about learned gene identity, since they are the sequence control, not a model arm, for any vocabulary-level question.

Construct caveat, stated because it is load-bearing. There is no cell vector in any of these models. Every "cell representation" is a mean-pool over gene-token residuals at one layer, i.e. approximately a context-corrected bag of genes. Part of the measured redundancy with raw expression is therefore built into the construction rather than discovered, and this should be borne in mind throughout § B.

A.2 Substrates

dataset	n	used for
Replogle K562 CRISPRi, non-targeting controls [9]	3,000 cells × 6,546 genes	cell cycle (primary)
Replogle RPE1, non-targeting controls	3,000	cross-cell-line frozen transfer
Setty CD34$^+$ human bone marrow [10]	5,780 × 14,319	development (workhorse); Palantir pseudotime, 10 clusters, 5 terminal branches
CZ CELLxGENE developing human gut, 6–11 PCW	4,269 (62,849 for co-expression panels)	development; the strongest co-occurrence panel
Tabula Sapiens lung airway [12]	3,600	development
Bastidas-Ponce mouse pancreas [11]	3,696	development; cross-species
Weinreb LARRY clonal barcoding [35]	865 undifferentiated / 167 clones	external ground-truth fate
Tabula Sapiens immune / lung / kidney [12]	˜7,500	cell type, gene context, co-expression baselines
`species_chrom.csv`	19,844 human genes	chromosome + genomic start

Phase coordinate. Tirosh S and G2M marker z-scores [13] → PCA(2) → φ = atan2, oriented G1 → S → G2M. Validated independently: phase-cluster circular means at G2M 20.3°, G1 150.4°, S 249.5° (correct cyclic order), concentrations 0.79–0.94, S and G2M score peaks in quadrature (109° ≈ 90°). Discrete composition G1 747 / S 1,208 / G2M 1,045.

Circularity, stated up front. φ and the phase call are *derived from marker expression*, so the expression baseline is circularly advantaged on every cell-cycle comparison. A model that beats it anyway is meaningfully better; an expression win is partly definitional. The cell cycle is a **degenerate testbed at the cell level** and an excellent one at the gene level, where the input can be stripped of the answer (§ C.2).

A.3 Two extraction defects found in this work

Input convention (§ 6.4). One extraction fed raw counts to a checkpoint configured with `input_style: "binned"`, `n_bins: 51`. Both versions were cached for three tissues, so the size of the error is directly measurable on the same cells:

metric	mean change when corrected	max	sign flips
pseudotime Spearman	**+0.109**	+0.214 (gut)	0
lineage balanced accuracy	**+0.118**	+0.277 (gut)	0
linear R^2	**+0.210**	+0.402 (gut)	0
curvature gap (kNN − linear)	**−0.075**	−0.118 (lung)	**1**
tangent rotation	−4.2°	17.0° (gut)	0

The direction is consistent: the defect **understates the model and overstates its curvature**. It compounds with a known confound (curvature significance tracks linear-decoder headroom at partial correlation −0.858), so raising linear R^2 by 0.21 mechanically shrinks apparent curvature. Nine routes rest on the uncorrected extraction.

Broken pseudotime. In the cached fetal-gut arm, mean pseudotime by cluster:

lowest		highest	
intestine goblet cell	0.18	progenitor cell	0.38
enteroendocrine cell	0.24	enterocyte	0.38
stem cell	**0.30**	colon epithelial cell	0.70

Differentiated cells sit below stem cells. The other three tissues are correctly rooted (blood HSC_1 0.11 → Ery_2 0.82; lung basal 0.24 → multiciliated 0.73; pancreas Ductal 0.23 → Alpha 0.78). Gut is excluded from the transfer analysis in § B.4 for this reason.

Other data hazards documented in the course of this work: `replogle_concat.h5ad` already stores log1p(CP10k), so re-normalising double-normalises and corrupts per-gene variance (it drove one gate from pass to fail); one cached activation file wrote local row numbers as global cell indices, producing a spurious "genuinely curved" result; one cell-cycle cache had no phase field and only a third of its cells matched the intended population; and C2S activation caches key rows **positionally** while every other cache keys by global cell ID, so a naive join yields 6 common cells out of 3,000 and silently produces garbage.

B. Cell-state geometry in full

B.1 The information decomposition and its two traps

Within-dataset, C2S-Scale on 3,000 K562 cells:

representation	phase ordering (circ-R^2)	classification
full expression, 6,546 genes with values	0.929	0.907
top-512 genes with values	0.885	0.853
top-512 set, binary, *the model's input*	0.882	0.846
top-512 set + rank, *the model's input*	0.868	0.844
C2S-2B L21	**0.875**	**0.866**
top-512 set + rank → MLP-256	0.802	0.830

Loss decomposition: 0.929 → 0.882 (tokenisation, −0.047) → 0.875 (the model, −0.007).

Cross-cell-line frozen transfer (K562 → RPE1). RPE1 selected on measured criteria over two rejected candidates: it carries 93/94 cell-cycle markers, is a single cell line so phase cannot confound with identity, and its phase is well spread (R = 0.267).

arm	R_{diff}	median error
all 6,544 shared genes + magnitudes	0.878	16°
top-512 + magnitudes	0.810	19°
top-512, rank only, matched gene selection, the model's input	**0.792**	20°
C2S-2B L21	**0.789**	23°

arm	R_{diff}	median error
constant predictor / uniform random	0.043 / 0.017	86° / 88°

paired contrast (5,000 draws)	$\Delta\ R_{\text{diff}}$	95% CI
model − matched input baseline	**−0.0022**	**[−0.0144, +0.0105] → parity**
tokenisation loss	+0.1042	[+0.0917, +0.1169]
...of which truncation to 512 genes	+0.0683	[+0.0576, +0.0787]
...of which discarding magnitudes	+0.0359	[+0.0261, +0.0459]

Two traps, each producing a different wrong verdict. Comparing the model against all 6,544 genes with magnitudes gives "expression wins", which charges the model for its tokeniser. Matching the *encoding* but not the *selection* gives "the model adds information" ($\Delta = +0.0153$, CI excluding zero), because a cell sentence draws its top-512 from the cell's **full** panel (RPE1: 8,749 genes) while that baseline drew from the 6,544 shared, so the model saw ˜2,205 genes the baseline never did. **Both the encoding and the gene-selection procedure must be matched.**

B.2 Downstream benchmarks

Five prior models, 1 of 45. Protocol: matched-dimensionality PCA ($k \in \{10, 20, 50\}$), small probe, held-out, three tasks. Expression wins ordering 15/15, classification 14/15, S-vs-G2M AUROC 15/15.

task (k = 50)	best model	expression	winner
phase ordering	MaxToki 0.924	**0.932**	expression
phase classification	**MaxToki 0.896**	0.879	model (the only cell of 45)
S-vs-G2M AUROC	MaxToki/STATE 0.991	**0.992**	expression

UCE loses all 9 of its cells by the widest margins (ordering 0.77–0.80 vs 0.93).

C2S-Scale, 0 of 36. Identical protocol and substrate. Best C2S ordering 0.876 (L25, k = 50) against expression 0.929, and slightly *worse* than the encoder models.

Developmental pseudotime, 0 of 20. Model + ridge(k = 400) minus the best nonlinear expression baseline, all negative:

	scGPT	Geneformer	STATE-SE	MaxToki	UCE
blood	−0.034	−0.063	−0.119	−0.026	−0.139
lung	−0.013	−0.041	−0.045	−0.013	−0.104
gut	−0.035	−0.134	−0.091	−0.021	−0.374
pancreas	−0.013	−0.032	−0.022	−0.015	−0.056

Expression + nonlinear ceilings: 0.977 / 0.983 / 0.921 / 0.987. A symmetric version (model given the same nonlinear probe) is also 0/20. Against a *linear* expression probe at $k \geq 100$ the models win 66/180, which an earlier $k \leq 50$ cap concealed; that correction is recorded in § F.

Steering, constrained advance at $\tau = 1.3$:

tissue	expr-PCA	scGPT	Geneformer	STATE	MaxToki	models beating expr
blood	**2.092**	1.446	0.626	1.898	0.382	**0/4**
mouse pancreas	1.728	2.152	1.426	**2.145**	2.087	3/4
lung	0.175	0.252	1.997	**2.270**	0.302	4/4
gut	0.769	0.877	0.996	**1.884**	0.642	3/4

Robustness: 5 seeds $\times$ 3 HVG settings, expression beats scGPT 15/15, Geneformer 15/15, MaxToki 15/15, STATE 11/15 = 56/60 on blood. **Do not read this as "models lose everywhere".** Blood is the tissue every earlier headline rested on, and it is the only one where expression wins outright.

B.3 No completion of deleted structure

Delete a contiguous arc of a circular trajectory from training entirely; score held-out gap cells with a readout fitted only on training cells. Two seeds, standard transformer, chance $R_{\text{diff}} \approx 0.03$:

gap removed	model	raw expression	model − data
0° (control)	0.993	0.998	−0.005
30°	0.994	0.998	−0.004
60°	0.991	0.997	−0.007
90°	0.989	0.996	−0.007
120°	0.963	0.990	**−0.027**

Both arms bridge remarkably well (removing a *third* of the ring costs the model only 0.030), because a ring is locally near-linear. But the model never beats the data, and its deficit grows with the hole. Bridge ratio (chord across the gap $\div$ mean arc step of an equal span) falls in both arms and converges at wide gaps (120°: model 0.28, data 0.35).

This matches the one real-data test, in which deleting an intermediate population and steering toward it produces the wrong lineage (held-out peak r = +0.545, rank 5/10, path running through the wrong branch), and adds the curve behind it: **steering interpolates; it does not generate.**

B.4 Cross-tissue transfer

Fit a pseudotime probe on one tissue, apply it **unchanged** to another. PCA fitted on the train tissue only. 10,529 genes shared across four tissues; every arm reduced to 50 components.

Within tissue (5-fold CV Spearman), the ceiling each arm transfers from:

arm	blood	gut	lung	pancreas
scGPT	0.945	0.845	0.948	0.916
expression	0.928	0.820	0.935	0.913
STATE	0.914	0.837	0.920	0.908
Geneformer	0.933	0.733	0.887	0.891
UCE	0.885	0.594	0.876	0.873

Across tissues, excluding gut (§ A.3), 6 ordered pairs:

pair	expression	scGPT
blood → lung	−0.208	−0.023
blood → pancreas	−0.205	**+0.589**
lung → blood	−0.337	−0.108
lung → pancreas	**+0.569**	+0.409
pancreas → blood	−0.305	−0.083
pancreas → lung	**+0.629**	+0.491

arm	mean	vs expression	wins	Wilcoxon
scGPT	+0.212	+0.188	4/6	p = 0.156
STATE	+0.138	+0.114	5/6	p = 0.156
Geneformer	+0.103	+0.079	4/6	p = 0.438
UCE	+0.087	+0.063	4/6	p = 0.562

Transfer largely fails for every representation. Within-tissue is 0.87–0.95; the best cross-tissue number anywhere is 0.63; only 3 of 12 pairs clear 0.5 and two are won by raw expression.

And the apparent model advantage is sign, not structure. Over all 12 pairs scGPT's signed advantage is +0.278 (Wilcoxon p = 0.016), but under **orientation-invariant** $|\rho|$ it falls to +0.094 (p = 0.233) and every other model goes to ≈ 0. Expression transfers with an *inverted* sign in 8 of 12 pairs, worse than useless; scGPT is merely closer to zero. The honest statement is that scGPT resists catastrophic negative transfer, not that it transfers well.

B.5 Where pretraining does pay

Endpoint targeting on hard trajectories. Steer a held-out root cell toward a terminal branch and ask whether its nearest real training cell is in that branch:

tissue	expr	scGPT	Geneformer	STATE	MaxToki	best model gain
blood	0.551	0.438	0.568	0.567	0.386	+0.017 (2/4)
mouse pancreas	**0.207**	0.539	0.365	0.458	**0.569**	**+0.363 (4/4)**
lung	0.599	0.652	**0.862**	0.833	0.729	**+0.263 (4/4)**
gut	0.728	0.669	0.661	**0.764**	0.711	+0.037 (1/4)

The pancreas gain is concentrated in the **rare** fates (expression delta/epsilon 0.010/0.010 against MaxToki 0.330/0.373), which is where a pretrained prior would be expected to help.

Linear readability. Pretraining makes trajectories linearly accessible: model + *linear* probe at $k \geq 100$ wins 66/180 (scGPT 22/36, MaxToki 25/36), a genuine win that dissolves entirely against a nonlinear probe.

A pretrained readout that names the fate. Fate-aimed steering read through the model's own head gives a clean diagonal: 14/15 target → fate hits across four tissues, pooled exact permutation over 103,680 assignments $p = 1.9 \times 10^{-5}$; MaxToki's untied `lm_head` reproduces 5/5 independently (D = +0.821, p = 0.0083).

B.6 Full anatomy tables

Geometry across representations (K562, 3,000 cells):

representation	linear circ-R^2	H_{flat} (planar?)	θ_{far} vs flat null
scGPT L11	0.892	not rejected (p = 0.905)	$37.5° < 54.2°$
Geneformer L11	0.844	not rejected	$59.6° < 63.3°$
MaxToki L8	0.918	not rejected	$35.2° < 53.1°$
STATE-SE L11	0.894	not rejected	$28.6° < 48.6°$
raw expression	**0.929**	not rejected	n/a
C2S-2B L21	0.875	n/a	n/a
C2S-27B L44	0.876	n/a	n/a

Depth construction in C2S (cross-validated circular correlation; chance = 90° error):

relative depth	C2S-2B	C2S-27B
0.00	0.068 (81°)	0.025 (83°)
˜0.25	0.321 (60°)	0.225 (67°)
˜0.35	**0.737** (26°)	0.581 (38°)
˜0.58	0.847 (17°)	**0.880** (14°)
˜0.80	0.898 (12°, z = +44)	0.891 (13°)
˜0.96	0.894 (12°)	**0.902** (13°)

Phase is near-absent at the embedding and constructed progressively; the 2B builds it faster in relative depth and both finish equal. *Caveat:* read at the final token, so early-layer values partly reflect propagation to that position rather than absence of computation.

Continuity battery (0 = continuous, 1 = discrete):

test	statistic	score
behavioural interpolation	largest-step share 0.257 (gradual 0.10, snap 1.00); mid-interpolation entropy +0.0056	**0.17**
metric profile	largest step 0.172 of total rise; monotone in 82% of bins	**0.10**
occupancy	decoded entropy 0.976 vs true 0.952; 0 empty bins	**0.00**

No activation plateaus. In language models, interpolating between two input sequences leaves the output nearly flat for most of the path and then changes sharply part way through; those plateaus sharpen with depth and track the input-output sensitivity of the MLPs [16]. Running the same interpolation here: transition width ≈ 0.5 of the λ range, where a plateau would be ≈ 0.05; plateau strength *decreases* with depth ($0.179 \to 0.115 \to 0.056$ at L09/L15/L21), the opposite of the reported depth trend; and crossing a real phase transition is indistinguishable from staying within a phase (0.068 vs 0.056). Proposed falsifiable account: a plateau needs a **discrete output alphabet**; a graded distribution over hundreds of marker genes has no cliff to form one.

C. Vocabulary-level facts in full

C.1 Chromosome

Decoding across representations (22-way balanced accuracy, common gene set n = 14,671):

basis	balanced acc	× chance	beats co-expression **and** sequence?
MaxToki `lm_head`	**0.483**	10.6×	yes
MaxToki `embed_tokens`	0.488	10.7×	yes
Geneformer V2	0.186	4.1×	yes
Geneformer V1	0.137	3.0×	yes
scGPT	0.089	2.0×	**no**
ESM-2 (sequence)	0.105	2.3×	reference
co-expression	0.085	1.9×	reference

(This table uses a weak linear probe on the common gene set; the headline 0.880 in § 4.1 is the 1B with its best probe on 15,135 matched genes under the 10-Mb holdout.)

The 10-Mb neighbourhood holdout, the decisive control:

basis	random split	**group split (10 Mb)**	retained
MaxToki `lm_head`	0.433	**0.347**	**78%**
MaxToki `embed_tokens`	0.433	0.373	85%
Geneformer	0.174	0.111	51%
scGPT	0.090	0.056	24%
ESM-2	0.190	**0.074**	**20%, collapses**
co-expression (developmental panel)	0.054	0.043	n/a

Why sequence predicts chromosome at all, and why it fails on HOX. Gene families arise by *local* duplication, so a gene's nearest ESM-2 neighbour sits on the same chromosome 22.3% of the time against 5.6% chance. HOX is the natural experiment that decorrelates sequence from locus: HOX arose by whole-genome duplication onto four chromosomes, so a HOX gene's nearest protein relative is its paralogue *elsewhere* (same-chromosome only 7.7%), and ESM-2 is at floor (0.023). **One biological fact predicts both the positive and the negative.**

Tokeniser confound, and how the first control was wrong. The vocabulary is ordered by Ensembl accession, and accessions are assigned in chromosome blocks, so adjacent token IDs share a chromosome 48.9% of the time against 5.6% chance. A *linear* probe on token ID returns chance, which is **false reassurance**, because the block structure is nonlinear: token ID alone via kNN predicts chromosome at 0.49 (k = 1). The finding survives the confound removed properly: holding out whole 1,000-token accession blocks retains **96%** (0.433 → 0.415). And the clean dissociation: both tables re-encode token ID about equally (embedding→token-ID $\rho = 0.46$ MaxToki, 0.44 Geneformer) yet score 0.433 and 0.174 on chromosome. If chromosome were re-encoded token ID they would match.

Layer profile (217M, 5,686 well-sampled genes): 0.453, 0.212, 0.185, 0.173, 0.168, 0.158, 0.146, 0.139, 0.122, 0.115, 0.098, **0.088** across L0–L11. Output table 0.516; input table 0.453. **1B at matched taps and matched width (k = 256):** 0.813 → 0.220 → 0.091 → 0.066 at L0/L2/L4/L8 (null 0.046), retaining **8%**; the 217M at the same taps gives 0.231 → 0.070. The 1B's advantage is concentrated at layer 0 (3.5× ahead) and is gone by layer 8.

Width is a confound. Reducing the 1B from 2,304 to 256 components costs almost nothing at L0 (0.910 → 0.813) but collapses L08 (0.265 → 0.066): the deep-layer signal is spread thinly rather than concentrated. Any native-width cross-model comparison measures width as much as content.

Causal steering, both models:

model	mean specific effect	chromosomes positive	random push
217M	+0.056	116/132	≈ 0
1B	**+0.170**	**132/132: every chromosome, every strength, both seeds**	≈ 0

Do **not** read the 3× as "the 1B uses chromosome 3× more": α is in units of each model's own mean gene-embedding norm and those differ (1B 3.52 vs 217M 0.73). What is not scale-confounded is the *consistency*: 132/132 against 116/132.

Depth of the causal effect: +0.0801 at the input embedding (22/22 chromosomes), ~26× weaker one layer in, −0.00058 with a CI spanning zero by layer 5 (11/22), exactly zero at the last layer.

Karyotype validation. The same directions reproduce K562's measured chromosome-level expression signature at r = +0.434 (z = +2.40) against a shuffled-karyotype control at ≈ 0.

Attention: measured, and negative. Same-chromosome versus different-chromosome attention, matched on rank distance (essential: the model uses RoPE, genes are ordered by expression rank, and same-chromosome genes co-express and so land at similar ranks):

sample	strongest head	heads with $\lvert z \rvert > 3$
6 cells	L2H5, z = **−1.13**	**0 / 88**
40 cells	L2H4, z = **+0.53**	**0 / 88**

The strongest head changes identity and flips sign between samples. **Attention is not sorted by chromosome at any layer or head.** The causal effect is real; the stated mechanism is not supported. *(217M only; the cell cycle × attention cell remains empty.)*

What does not hold. Sub-chromosomal position is decodable (ρ +0.396, 22/22 chromosomes, beating both baselines) but the factorisation beats **both** models on it (ρ +0.762 vs +0.622 and +0.374), so position-beyond-co-occurrence does not survive; and the causal response is **flat** along a chromosome, so fine position is readable but not used. Steering destinations do not replicate across models (3/22 modal agreement) and fail against expression enrichment in both (p = 0.92 and 0.115), so the destinations are model-specific geometry, not biology. The one positive mechanism result, 5-Mb local domains at +0.049 and +0.037, both $p \leq 0.0001$, is a *characteristic scale*: at ~150 Mb the correspondence is undetectable and at 2 Mb it weakens.

Chromosome is entangled with cell identity. Chromosome steering disturbs cell-type, lineage, tissue and cycling readouts more than a matched meaningless push, and *more* in the 1B (cell-type disturbance +0.541 vs +0.208). It is not stored in a subspace separate from cell identity.

C.2 Gene identity → cell-cycle phase

Design: a real cell with all 99 cell-cycle genes stripped (phase-neutral backbone), plus one gene symbol at a fixed rank. Readout: which pole of the real-cell phase circle the last-token state lands on. Full surface-form ladder in § 4.2.

Constructive sufficiency (the user-designed variant). Insert 8 genes whose **external** Cyclebase/Whitfield [14] peak-phase annotation (a synchronised time-course measurement, not this dataset's covariance) is nearest a target angle:

condition	angular separation
annotated (true phase assignment)	**114.3° ± 6.9**
shuffled, *same 43 genes, permuted labels*	13.2° ± 3.7

condition	angular separation
random, 8 non-cycle genes	3.5° ± 0.8
real cells of those phases	*175.7°*

≈33σ against random and ≈10σ against shuffled: the movement is driven by **which phase the genes belong to**, not by cell-cycle gene content. Radius behaves as the disk reading predicts: annotated modules stay at real-cell radius (0.752) while incoherent mixtures collapse toward the centre (0.508). At 27B the effect grows with depth (43.5° → 69.9° → 96.8°) with flat controls.

Limits. Only ˜65% of the real separation; 80–116° absolute offset (8 genes ≪ a full programme); poor within-arc resolution. **Loop closure is untestable** from annotation alone, because ˜40% of the circle (≈270–330°) has no phase-specific transcripts, so those targets return the 0° module, and a `loop_gap = 0 deg` printed by the harness is an artefact, not closure.

Compute-versus-relay. Deleting all 87 markers from the input retains 95% of phase decodability, but a matched baseline retains 0.94 under the identical ablation, so this is **not model-specific**. The cell cycle modulates transcription genome-wide; this rules out "copies the markers", not "reads a distributed signature".

C.3 Gene context, function and switches

Genes are re-referenced by context, early, then it decays. Crowd-removed pairwise context-shift agreement minus a gene-shuffled control, MaxToki-217M: L0 **+0.000 exactly** (sanity), L1–2 peak **+0.834**, monotone decay to L11 +0.63. Different-gene control +0.0008. Rank-residualised retains 96%. No dimension carries more than 6.6% of variance.

Architecture differs by ˜10×. EXCESS (gene-specific context response) and func-z (organisation along a nuclear-vs-surface axis against a random-axis null) at L4: scGPT 0.705 / +2.74; STATE-SE 0.835 / **+1.31**; MaxToki-217M 0.740 / **+21.17**; MaxToki-1B 0.881 / +16.43. STATE-SE is the striking case: it moves genes hardest in the *least coherent* direction, because its gene tokens are frozen ESM-2 vectors carrying sequence rather than co-expression.

How much of contextualisation is learned? An untrained, random-init model of identical architecture and vocabulary gives EXCESS **+0.274** (against +0.74–0.76 trained) and func-z **+3.8** (against +21.2). So there is a real **architectural floor** of about a third, since random embeddings plus deterministic attention mixing already produce a reproducible gene-specific shift, but the **functional organisation is almost entirely learned**.

The functional-context axis is causally used, but it is not model-specific. Inject a nuclear↔surface axis built at L4 into half a cell's gene tokens after L3, read native next-gene logits on GO sets at the *other* half: signed swing **+0.8149**, 28/30 cells, $\mathbf{p} = 4.3 \times 10^{-7}$, dose-monotone (+0.088 / +0.271 / +0.815 at $\alpha = 0.25/0.5/1.0$), norm-matched random ≈ −0.01. **But the axis fails its co-expression-coherence null at +1.16σ.** This is the cleanest single example of "causally used but not the model's own".

Antipodal regulator pairs. RORC/FOXP3 scGPT $p < 10^{-4}$, separation 0.476; GATA1/SPI1 $p < 10^{-4}$, separation 0.818, subspace cosine −0.394/−0.466/−0.195 across models against **+0.007 co-expression** and **+0.699 ESM-2**; PAX5/PRDM1 $p \leq 0.0005$ in all three models. TBX21/GATA3 and SOX2/CDX2 are null everywhere. Beats both baselines on 3 of 5 pairs.

Steerable bistable switches. GATA1/SPI1 **+4.211** [+3.892, +4.539], reciprocal (ΔR_A +1.830, ΔR_B −2.175); PAX5/PRDM1 **+5.565** [+5.225, +5.902]. Off-tissue RORC/FOXP3 reproducibly *anti* (−1.9 / −1.6). Norm-matched random push ≈ 0. **No expression or sequence baseline arm was run for this result**: it gates on steerability, not novelty.

Paralogy. Leave-one-cluster-out held-out ρ: A +0.891, B +0.709, C +0.833, D +0.900; mean **+0.833**, null −0.000 ± 0.231, z = +3.61, 4/4 signs (p = 0.005). ESM-2 **+0.889**; co-expression (Tabula Sapiens) +0.086 (n.s.); co-expression (fetal gut) +0.506. **Beats co-expression, loses to sequence.** The analogy `HOXA9 - HOXA1 + HOXB1 -> HOXB9`: top-1 0.333, median rank 2 of 20,271, **3.95× a strict within-cluster**

null (z = +13.5); co-expression 1.3× (p = 0.333, cannot do it); ESM-2 10.7×; the ranking *inverts* between loose and strict nulls.

Gene function beyond co-expression (C2S). Within (co-expression × mean-expression) bins, Δcos same-function vs different: L9 0.0619 (null −0.0009 ± 0.0095, **z = 6.57**, p = 0.005); L13 0.0439 (z = 3.86); L17 0.0547 (z = 4.52). Caveat: one tissue, 3–6 bins.

The shape of the chromosome object. It is a **clustering**, not a manifold: four HOX clusters separate at held-out 4-class accuracy 0.974 (1B) / 0.923 (217M), but the axis is invariant to random re-coding of the labels and inter-cluster offsets have mean cosine ≈ 0.2 against ≈ 1.0 for a real grid. Causally it behaves as **22 independent, signed, saturating lookups consumed at the input**, not one continuous genomic manifold. The *paralog* axis, by contrast, is a genuine shared 1-D coordinate that composes across clusters.

HOX causal use is cluster-specific and tissue-invariant. $\beta = +0.093$ (fetal gut) and +0.145 (bone marrow), both t > 6, placebo-clean, replicating across two independent tissues. Per cluster: HOXB +0.26 / +0.25 (used in both tissues), HOXA −0.03 / +0.055 (null in both, *even where it dominates*). A rotation test, moving to a tissue where HOXA dominates, **falsified** the tissue-gating interpretation: the used cluster does not follow the dominant one. The live hypothesis, untested, is that HOXB is the most consistently corpus-wide co-regulated cluster.

D. Synthetic experiments in full

D.1 Design

Corpora are generated from a latent structure we choose, pushed through scGPT's **published** input encoding (51 quantile bins, expression-sorted truncation), into a **plain standard transformer**: softmax multi-head attention, GELU feed-forward, pre-norm, d = 192, 4 layers, 4 heads. Nothing about the model is novel. 20,000 training cells, 1,000 genes, 3,000 steps, `val_corr` 0.71–0.86 throughout.

Why an unremarkable model. These experiments need many corpora, not one large model; and a bespoke architecture would let any result be attributed to its unusual components rather than to transformers in general.

D.2 The metric-warp law

Plant a circle with an even metric; break one thing inside a 90° arc. Statistic = (model stretch − data stretch), so only a model-specific effect registers. Stretch = mean knot gap inside the arc ÷ mean gap outside; centroids computed on an equal number of cells per bin so density cannot drive centroid noise.

arm	model stretch	data stretch	**model − data**	seeds > 0	vs control
uniform (control)	1.007	0.992	+0.027 ± 0.095	4/8	n/a
sharp (output turns over fastest)	1.202	1.032	**+0.187 ± 0.091**	**8/8**	**+0.160, t = 3.43, p = 0.0041**
occupancy (3× cells)	0.842	0.774	+0.097 ± 0.039	7/8	+0.070, n.s.
noisy (harder to predict)	1.040	1.139	**−0.161 ± 0.080**	**0/8**	**negative**

Localisation, seeds 3–7 (manipulated arc = bins 5–7):

arm	argmax bin per seed
sharp	**5, 5, 6, 6, 5**, inside the arc every time
uniform	1, 10, 5, 4, 2, scattered
occupancy	10, 8, 11, 1, 11, mostly outside
noisy	4, 7, 4, 4, 7, clustered at the arc edge

Caveat. Global CV excess over each representation's own shuffle floor is ≈ 0 or negative in three of four arms. The `sharp` effect is a **localisation** result, meaning the stretch sits where the manipulation is, rather than a claim that the model's loop is globally more uneven than the data's.

D.3 The token table

Assign every gene an arbitrary group label appearing in **no single cell**; sweep how strongly group membership drives co-occurrence. 22-way probe on the table, split over genes, chance 0.050. PPMI is a training-free factorisation of the identical corpus.

consistency	model table	PPMI	margin
0.0	0.047	0.061	−0.014
0.2	0.160	0.202	−0.042
0.4	0.245	0.290	−0.044
0.6	0.296	0.376	−0.080
0.8	0.340	0.440	−0.100
1.0	**0.426**	0.497	−0.071

Generalisation to genes never seen together. Split every group's genes into halves and let programs draw from one half only, so within-group *cross-half* pairs never co-occur, the synthetic form of the neighbourhood holdout. 3 seeds:

	observed pairs	held-out pairs	combined
model table	0.1367 ± 0.0068	$\mathbf{+0.0147 \pm 0.0107}$	**Stouffer z +20.2**
PPMI	0.1946 ± 0.0077	$+0.0037 \pm 0.0011$	n/a

The model generalises to unseen pairs at **four times** the factorisation's own rate.

The table is load-bearing:

`gene_emb`	`val_corr`	vs real
real	0.8534	n/a
zeroed	0.0842	−0.769
row-shuffled	0.0047	−0.849
frozen at random init	**0.6118**	**−0.242**

Exclusions ruled out. Not task degeneracy (a per-gene-mean predictor scores 0.47 against the model's 0.85). Not a baseline mismatch (PPMI restricted to the model's own top-128 input still scores 0.453 at consistency 1.0 against 0.491 on all genes). Not undertraining (1500 → 12000 steps moves the probe by 0.003 while `val_corr` rises 0.801 → 0.857; note the trap: `corr(log steps, acc) = +0.96`, perfectly monotone and entirely inside chance). Not probe linearity.

D.4 The steering arm, and why it is excluded

The group direction is built from real embedding rows, `mean(W_E[group == c]) - mean(W_E)`. Three successive nulls failed:

1. **Norm-matched isotropic random.** At consistency 0.0, where the true effect is exactly zero, this reads −0.98 mean with 38% of groups positive, reaching −7.1 at consistency 1.0. Cause: any direction built by averaging real embedding rows sits in the dominant part of an anisotropic embedding distribution.
2. **Construction-matched** (same arithmetic, shuffled labels). Calibration point still −0.28.
3. **Alignment-matched** (each null draw carries its own shuffled labelling used for *both* the direction and the readout, each deflated by its own no-push baseline). Calibration point +0.097 / −0.029 / −0.269 across doses, fine at low push but drifting with dose.

The null-versus-null floor settled it. Scoring one structureless labelling against another with identical machinery:

α	null-vs-null floor	"real" reading (also structureless)
0.5	−0.105	+0.039
1.0	−0.276	−0.002
2.0	**−0.489**	−0.227

The floor exceeds the signal at every dose. The bias is structural: the real arm is a *single* labelling while the null is the *mean of five*, and with a saturating dose response those are not exchangeable, which is why the bias grows with α. **No steering claim is reportable from this experiment.** The synthetic work therefore establishes readability and beyond-baseline for the token table and is **silent on causal use** there. The real-model causal evidence (§ C.1) is a separate measurement with its own controls.

D.5 The stall theorem, verified

Synthetic circle lying exactly in a linear 2-plane of a 60-D space (flat by construction):

arm	peak advance	laps	off-manifold	$\|D\|$ min	reversible
fixed	+1.50	0.24	11.08	1.00	−1.67
fixed + projection	**+1.49** (predicted $\pi/2$ = **1.571**)	0.24	1.08	**0.09**	−1.70
fixed + projection + retraction	+1.50	**0.24, no rescue**	0.75	0.64	−1.57
local + projection	+10.76	1.71	1.65	0.66	−10.84
local + projection + retraction	**+31.09**	**4.95**	0.68	0.92	−31.06
transport (label-free)	+2.45	0.39	1.44	**1.00**	−2.12
oracle (uses labels)	+24.05	3.83	0.64	0.27	−0.94

The $\|D\|$ **collapse to 0.09** is a judge-free measurement of w $\perp$ $\hat{t}$: the raw pre-normalisation step norm *is* $|\mathrm{w}{\cdot}\hat{t}|$.

In seven real representations:

representation	fixed + proj	+ retraction	local + proj + retraction
scGPT L11	0.34 laps (dies T24)	0.38 ✗	**4.53**
Geneformer L11	0.14 (T13)	0.18 ✗	**5.57**
MaxToki L8	0.01 (T1)	0.02 ✗	**6.03**
STATE-SE L11	0.31 (T22)	0.29 ✗	**4.70**
raw expression	0.36 (T40)	0.34 ✗	**5.08**
C2S-2B L21	0.216 ± 0.170	0.159 ✗	**3.452 ± 0.129**

`retraction_rescues_fixed = false` in every substrate. **Honest negative:** the label-free `transport` arm fails everywhere (0.15 laps, $\|D\|$ collapse ratio exactly 1.00, since the carried direction never rotates), so there is currently **no label-free version of the working operator**.

D.6 The operator ladder

Mean over 16 model × tissue cells; every rule integrated at identical L2 path length; advance judged by an independent kNN regressor fitted on training cells only.

operator	CA@1.3	tangent alignment	off-manifold @T=8	verdict
linear (constant direction)	0.461	0.420	1.847	baseline
quadratic (global bilinear)	0.426	0.400	˜1.85	**refuted 16/16**
local (no projection)	0.389	0.402	1.864	worse in 10/16
piecewise-5 / piecewise-10	0.288 / 0.228	0.486 / 0.540	1.675 / 1.658	monotonically worse
linear + projection	**1.239**	0.692	1.355	**+0.778, 16/16**
local + projection	**1.320**	0.683	1.347	best; beats oracle 13/16
oracle (uses labels at inference)	0.676	0.774	0.686	not a competitor

Ablations: projection alone +0.778; local direction alone −0.072. **On a closed loop the balance reverses**: gain from locality +2.09 (scGPT), +4.66 (MaxToki) against gain from projection +1.45 and +0.06. The general operator is local direction + projection + retraction, and topology decides which term matters.

Why the global quadratic failed, measured rather than inferred: Q's own gradient field rotates only 12–96° early→late against a measured true tangent rotation of 48–113°. One global quadratic under-rotates by roughly half and cannot express the direction discontinuity at a bifurcation.

The direction/projection 2×2. The original `random_dir` control changed two things at once. Replaced by {developmental, random} × {projected, not}: read through scGPT's own MVC head, dev-axis cosine +0.0637 (random + projection) vs +0.3485 (real direction + projection); differentiation contrast +0.522 vs +2.155. **The projection alone steers nothing; the direction carries the biology.**

E. A worked negative: the extracted-operator battery

An earlier submitted result claimed a compact developmental representation "extracted" from a frozen model: an operator exported from the model, a low-dimensional adaptor fitted to a curated stage ontology, and task-specific readouts, benchmarked against scVI [23], Palantir [10], DPT [24], CellTypist [25], PCA, SVD and raw expression across 88 donor-holdout splits with BH-corrected Wilcoxon tests. Roughly 25 control arms were then added. It does not survive them.

What the operator actually is. Not a weight matrix. It is a sum of post-softmax attention probabilities accumulated over 600 cells, never divided by its companion counts file, so row sums equal per-gene detection counts exactly ($r = 1.0000$), and 580 of 1,200 genes are undetected in that corpus, so every operator has 580 identically-zero rows. The model in question is $d = 512$ with 8 heads, so no parameter tensor is $1{,}200 \times 1{,}200$; that is the sequence length. "Direct export from frozen weights" is therefore false: the stage requires a forward pass over target-domain cells.

It ties a random matrix. At the cell level there is **no bottleneck width $\geq$ 10 at which the real operator separates from a matched Gaussian on the same support**. At $d = 10$ two identically constructed random nulls differ from each other by *more* than the real operator differs from them (null-vs-null $p = 0.016$ against the claim's $p = 0.156$). At full capacity all three operators tie within 0.003.

Training budget inverted the ranking. At the paper's own 220 epochs the real operator is beaten by a Gaussian (0.711 vs 0.852), by co-expression (0.762) and by its own shuffled values (0.758), and all 96 heads sit below the worst of 20 random draws. At 2,000 epochs the real operator leads, by +0.009 over its own shuffle, +0.011–0.015 over co-expression and +0.027 over a Gaussian, against a null-vs-null floor of 0.006. **A fixed epoch budget is not a neutral choice.**

What the selection score measures. Across all 192 circuits the head-selection score correlates **$r = 0.088$** with the biological endpoint it was chosen for, and $-$**0.90** with cell-level stage accuracy. Destroying gene-pair identity while preserving the value distribution on the same support reproduces 49–80% of the real operator's margin; preserving the exact spectrum while permuting the gene axis reproduces little. So the score measures "were the genes mixed by a reasonably conditioned dense matrix", not by *which* matrix. A ˜0.42 gap opens between no-operator (0.44) and any operator (0.77–0.81), and everything after sits in a ˜0.07 band that the null-vs-null floor (0.042) nearly fills.

Weight-derived operators do not rescue it. The best is the pure learned gene-embedding gram at 0.8634, against its correct matched null (the same embedding with gene rows permuted, preserving geometry and spectrum exactly) at 0.8477. The gap is 1.0 sd of the null-vs-null floor. Every pooled QK and OV operator sits at or below that null. And there is an identity behind it: $x \cdot EE^{\mathsf{T}}$ is a rank-512 linear reparametrisation of `x`, so a full-capacity linear probe cannot see anything in it it could not see in raw `x`.

The model's own representations lose to a random projection. On CD4/CD8 subtype discrimination across 14 held-out donors, a dimension-matched **random projection of raw genes** beats the model's frozen embeddings by **+0.0507 (14/14 donors, $p = 0.0001$)**; PCA-512 by +0.0472; raw genes by +0.0424. L2 normalisation is not the cause (the unnormalised vectors have norm CV 1.03%, and re-extraction with normalisation off changes the result by +0.00003, $p = 0.95$); nor is the readout choice (`<cls>` vs avg-pool: +0.0061, $p = 0.36$, still 0.043 below the random projection).

One incidental finding. That model's preprocessing breaks tied quantile bins with an unseeded global RNG, so its embeddings are not reproducible run to run: the same cells re-extracted agree at cosine 0.9896, while agreement with the cached version is 0.9901, so the pipeline agrees with itself no better than with a different run.

What is left standing: Stage-1 features outperform raw expression by 0.051 AUROC on T-cell subtype discrimination across 14 held-out donors. But a random matrix on the same support performs equivalently, so the gain cannot be attributed to learned structure.

We report this in full because every individual check the original performed was passed. The failure lies in the choice of checks: none of them was a matched random operator or an information-matched baseline.

F. Retractions, instrument failures, and things that did not work

F.1 Claims we withdrew

withdrawn	replaced by
"Chromosome *and* within-chromosome position beat co-expression"	The control was a raw per-gene expression profile, which sits at chance and excludes nothing. A proper factorisation of the identical cells reaches 0.720, which beats the 217M outright and takes *position* away from both models (ρ +0.762 vs +0.622 / +0.374). **Two claims retracted from one methodological error.**
"A global quadratic gradient field steers better than a constant direction"	Refuted 16/16. Best real gap +0.206, sign unstable, synthetic-linear control $\approx$ 0.
"The GATA1↔PU.1 saddle is a mutual-repression geometry"	Refuted twice: representationally (product R^2 -0.004 vs s^2_{ery} +0.114) and in the biology (correlation among undecided progenitors -0.050 against a -0.15 threshold).
"The cell-cycle manifold is a low-dimensional ring"	A **filled disk**; the earlier statistic was computed on 12 bin centroids and is void as single-cell dimensionality.
"C2S sharpens phase ˜3× over its own input"	The comparison used a `1/rank` weighting that collapsed the baseline's PCA onto a few genes. A faithful encoding gives 0.868–0.882, level with the model at 0.875.
"Compute, not relay: 95% retention after deleting markers"	Not model-specific: a matched baseline retains 0.94 under the identical ablation.
"Human curved, mouse flat"	Significance tracks linear-decoder headroom (partial corr -0.858 vs -0.091 for species).
"Steering the manifold is retired: there is no curved arc to ride"	*Global* steering fails; local-tangent projection works.
"Projection is the whole story"	True on trees, false on loops (§ D.6).
"16/180: models are useless on pseudotime"	An artefact of an inherited k $\leq$ 50 cap; 66/180 at k $\geq$ 100. The correct verdict is 0/20 against a *nonlinear* expression baseline.
Aging/HSC headline	Withdrawn by its own held-out test: disease donors sit inside the healthy range (p = 0.23) and the winning lineage flips between runs. What survives is a "real, small, and on current evidence useless" direction-vs-position signal.

F.2 Instruments that broke

- θ_{far} **must not be used on periodic manifolds.** On a circle the first and last bins are adjacent, so a perfect circle scores $\approx$ 0; the real loop scores *below* its own flat-circle null in every model.
- **The decodability gap (kNN − linear) is not a curvature statistic.** It is structurally headroom-limited and correlates -0.926 with linear R^2. Deprecated.
- **Circular correlation is degenerate against uniform variables.** It uses sin(a − circmean(a)), so a constant rotation cancels and the null equals the real value exactly; and it returns r = +0.804 at p = 0.0005 for an arm that travelled **zero** laps. Replaced by decoded winding and a time-shift null.
- **Full-space kNN breaks at high dimension** (curvature -0.457 at d = 2048).

- **A whitening bug** made the first nonlinear expression baseline return $R^2 = -2.583$, and **the broken version would have confirmed a model win**.
- **Permutation-floor p-values** (1/13, 1/21, 1/41, 1/201) were quoted as evidence in several places.
- **Small-n leave-one-out is degenerate:** pure noise scores $|\rho| = 0.736$ at n = 9.
- **A saturated test cannot distinguish present from absent:** perturbation identity decodes at 1.000 from a *random-init* model because the label is an injected input.
- **MPS boolean indexing** returned mismatched element counts for the same mask (3243 vs 40) at scale; removed from the hot path.

F.3 Nulls that were wrong, and the general lesson

Six documented cases where a large z against a permutation null went to ≈ 0 against a competitor built from the same cells: attention z $+4.18 \rightarrow +0.17$; the functional-context axis $+20\sigma$ random-partition, $+9\sigma$ tightness, **$+1.16\sigma$ co-expression**; paralogy beats co-expression and loses to sequence; C2S phase 0.898 against a broken input encoding and parity against a faithful one; cross-model consensus survives every hygiene attack until a matched co-expression neighbourhood scalar recovers ρ 0.458–0.581; and a label made of **pure sampling noise** scoring 0.447 at z = 35.8, higher than the real biological label at 0.357, z = 24.9.

F.4 Things that did not work

- **Perturbation inversion.** Negative in 4/4 models including the perturbation-trained one. AUROC 0.487–0.707; 0/4 fates significant; residualised on differential expression all q $\geq$ 0.26. A model trained on a CRISPRi vocabulary passes its validity gate yet ranks the canonical erythroid regulator 111/131 **with the sign backwards**. *Caveat:* the differential-expression-matched stratum test cannot run (0–2 known drivers fall in the low-|DE| half), so the negative is "no signal beyond DE", not "provably no regulatory knowledge".
- **Hidden clonal fate (LARRY [35]).** A well-powered negative: expression position AUROC 0.767, geometry 0.540 $\approx$ its clone-permutation null 0.383 $\pm$ 0.061, combined gain -0.006.
- **Reconstructing a deleted intermediate.** Held-out peak r = +0.545, rank 5/10, path running through the wrong lineage. Steering interpolates; it does not generate.
- **Commitment as a discrete geometric step.** No cliff (sigmoid transition width 0.27–0.83 of the pseudotime range), and a counts baseline wins in 3/4 models.
- **Label-free `transport` steering.** Fails in all five substrates, $\|D\|$ collapse ratio exactly 1.0000.
- **Irreducible topology.** No loops or holes anywhere. In-plane H1 null in five substrates including raw expression (all p $\geq$ 0.27); out-of-plane excess negative in 5/5; lineage-tree H0 z = -6.81 (branches *tighter* than a matched Gaussian).
- **Bilinear/interaction machinery generally.** Supervised bilinear CP probes lose to linear (-0.060); SAE dictionaries [36] are 73–99.7% redundant with linear ones and order-4 is worse than order-2; and third-order combinatorial ablation finds **0.00% superadditive interactions over ~2,980 targets**.
- **Scale.** 5× parameters changes rotation excess by $-11.1°$ and $\Delta\rho$ by 0.00; at matched SAE capacity a 13× larger model has 31× more dead features and 19 points lower annotation.

F.6 Open questions we consider most important

1. **A neighbourhood-co-regulation baseline for chromosome.** Every existing baseline is a per-gene profile or a global factorisation; neither models *local* co-regulation. Scoring against Hi-C A/B compartments and replication timing decides whether the strongest positive survives.
2. **What makes a token table a map rather than an index**: architecture, corpus scale, vocabulary size, or a separately-used output head.
3. **Cell cycle × attention**, the last empty cell in the coverage grid.
4. **A causal test for token-table facts under a calibrated null**, given § D.4.
5. **The output-change law on a non-cyclic manifold.**
6. **Correcting the two extraction defects** across all affected analyses (§ A.3).